PÓLYA INEQUALITIES FOR COMPACT PLANAR JUNCTION NETWORKS

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Abstract. We prove Pólya's Dirichlet inequality for a class of compact planar networks with unrestricted finite combinatorial complexity. Unit-square chambers are centered on a square lattice of spacing three and joined along selected nearest-neighbour edges by centered passages of length two and width one half. For every finite choice of vertices and edges, every eigenvalue satisfies $\lambda_k \geq 4\pi k/|\Omega|$. There is no periodicity, connectivity, or density requirement on the selected graph. Explicit connected examples have arbitrarily many holes and uniformly bounded diameter after area normalization; they do not tile the plane by congruent copies.

The proof reduces the global count to six mixed-boundary chamber comparisons. Physical Dirichlet segments raise the few levels at which the separated rectangular comparison is insufficient. Boundary moments control repeated eigenvalue clusters, including their infinite-dimensional complements. The four-port cell is certified by small interval matrices and an explicit additional harmonic witness. A phase-uniform fractional-part estimate gives an explicit high-energy bound; exact integer arithmetic then certifies the remaining finite interval. Explicit additional width–length ranges allow unequal passages and independently displaced windows at the same junction, with exact physical-area accounting. A complementary theorem treats chains of rectangular chambers with displaced ports. All estimates count eigenvalues with multiplicity and include equality at the spectral threshold.

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1. Introduction

For a bounded planar domain Ω, the eigenvalues of its Dirichlet Laplacian describe the squared frequencies of an ideal membrane fixed along its boundary. Weyl's law identifies their leading distribution: the number of eigenvalues below a large energy E is asymptotic to $|\Omega|E/(4\pi)$. Pólya's conjecture asks whether this leading expression is already an upper bound

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at every energy. Equivalently, if the eigenvalues are listed increasingly with multiplicity, the conjecture is

$$\lambda_k(\Omega) \geq \frac{4\pi k}{|\Omega|}, \qquad k \geq 1. \tag{1.1}$$

An asymptotic remainder, even one with the expected boundary sign, does not by itself settle the finite range of frequencies. That range becomes especially important when a domain contains many junctions: cutting the domain into elementary pieces simplifies the spectrum but also releases boundary constraints at every cut.

This paper treats networks in which that loss can be paid using the Dirichlet boundary that remains on each chamber. The chambers are unit squares centered at $3v$, with v in a finite subset of $\mathbb{Z}^2$. Selected nearest-neighbour pairs are joined by rectangular passages of length two and width $1/2$. Then (1.1) holds for every eigenvalue, including when chambers have four incident passages. The selection may contain missing vertices, missing edges, branches, cycles, or several connected components. In particular, the result is uniform in the number and arrangement of the pieces, rather than an eventual estimate for one fixed network.

The geometric scope is visible in the fully occupied q-by-q grid. For each $q \geq 2$ we obtain a connected domain of area $3q^2 - 2q$ with $(q-1)^2$ holes. After rescaling to area one its diameter is less than $\sqrt{6}$, independently of q. Each hole has area 6 before rescaling, whereas the domain has area at least 8; this gives a short obstruction to congruent plane tiling. Thus the examples can have increasingly complicated topology within a bounded normalized region, without obtaining the spectral bound from a tiling of the plane by the domain itself.

The fixed proportions are a starting point for a wider, explicitly specified family. Passages of different lengths and widths can meet at the same chamber, and their windows need not be centered. The admissible dimensions are given in Section 7; they include an unbounded continuous length range and substantial window displacements when the passage is narrower. The area in the bound is always the sum of the actual room and passage areas. The argument does not replace this sum by a common reference area or require additional large terminal rooms to supply the missing budget.

Position among all-eigenvalue estimates. Pólya's original argument establishes (1.1) for tiling domains by combining bracketing with a large-scale spectral limit [1]. Tiling and inheritance remain powerful tools: the Dirichlet or Neumann inequality passes from a known domain to a congruent subtile, as formulated in [2]. Non-tiling families are therefore an established part of the subject [4], and the all-eigenvalue Dirichlet results for Euclidean balls [2] and concentric planar annuli [3] show that sharp counting need not come from an explicit separated spectrum. Sufficiently thin Cartesian products provide a different class of all-eigenvalue results [6]. More recently, Jiang and Lin obtained quantitative strip-tiling estimates and sufficient conditions for removing collections of squares [5]. These results make the distinction between a new geometric class and a new general spectral principle particularly important.

The compact networks here are treated directly through their mixed-boundary cells. Their main contribution is the all-eigenvalue family theorem, together with a complete local certificate that is independent of the size of the network. The explicit fully occupied grid members cannot be congruent subtiles of plane-tiling parents, since such a partition would itself yield a plane tiling. They also fail the checked quantitative finite-square-removal criterion with any connected tiling parent of the specified kind; the area comparison is given in Section 2. These observations identify concrete distinctions from familiar sufficient criteria, without requiring a classification of all possible spectral inheritance constructions.

Rooms-and-passages domains and their mixed rectangular comparisons have a substantial history; see, for example, Brown, Evans and Wood [7]. Here the aim is an exact sharp count over the whole spectrum for a finite network with arbitrarily many pieces. The treatment is complementary to the square-hole inequalities in [13]. All comparisons and certificates needed for the network and chain results are proved below.

How the local comparison closes. After cutting at the chamber–passage interfaces and imposing Neumann conditions at the cuts, an edge contributes one passage spectrum and a vertex contributes a mixed chamber spectrum. Half of each edge contribution can be assigned to each endpoint. It therefore suffices to prove a local inequality for each of six port-incidence types: no port, one port, two opposite ports, two adjacent ports, three ports, and four ports.

Replacing an entire port-bearing face by a Neumann face makes the chamber separable. This comparison is already sufficient outside finitely many bounded frequency intervals. The exceptions cannot be dismissed by asymptotics: at certain low rectangular levels the count is genuinely larger than the allotted area term. The uncut Dirichlet portions of the faces provide the missing information. A trace estimate lifts a simple level, and two boundary moments lift a two-dimensional cluster in the three-port chamber. The latter estimate keeps the whole orthogonal complement through Parseval's identity and an explicit resolvent-weighted sum. For the four-port chamber, a finite-block version of the same argument yields certified counting bounds from matrices of order at most six. Every matrix calculation is enclosed by rational arithmetic, and the entire omitted spectral complement is bounded analytically.

These are quantitative applications of standard variational and intermediate-problem ideas, not new min–max principles; compare [10, 9]. Separated mixed spectra of component rectangles have also been used with Weinstein determinants for L-shaped, notched, and crossed regions [11]. Qualitative strict mixed-boundary monotonicity [8] would not supply the size of the shifts required here. Likewise, an ordinary conforming trial space gives upper eigenvalue bounds, whereas the present count requires lower bounds. The distinction is central to verified eigenvalue computation [12].

The rest of the spectrum is handled by a fractional-part argument. Consecutive rows in a lattice count cannot all have arbitrarily small fractional losses when their slopes traverse a suitable interval. A minimized block sum makes that statement uniform in the integer and half-integer shifts arising from the boundary conditions. It produces a fully explicit tail. Below the tail, the verification consists of integer event counts and rational inequalities, including every point at which a spectral correction is switched off. No numerically approximated mixed eigenvalue is used in the proof.

A comparison of two valid chamber budgets then permits heterogeneous passages: a low-frequency estimate controls an edge before its first transverse mode, while a rectangular counting estimate controls the remaining modes. Because the comparison is pointwise at the same energy, different edges can use different alternatives simultaneously. Window inclusion transfers the result to displaced, narrower openings without imposing symmetry on their physical eigenfunctions.

A further application retains independent displacements of the ports along a chain of 1-by-2 chambers and 4-by-1 passages. It illustrates how symmetry of a comparison cell can control physical domains without that symmetry. The paper concludes with a precise limitation of independent-cell cutting: the same method cannot remain uniform when a junction loses essentially all of its Dirichlet boundary.

2. The networks and their local spectral budget

2.1. **Geometry and the all-eigenvalue statement.** Let $V \subset \mathbb{Z}^2$ be finite and nonempty. Let F be any subset of the unordered nearest-neighbour pairs with endpoints in V, and write

$$d(v) := \#\{e \in F : v \in e\} \le 4 \qquad (v \in V). \tag{2.1}$$

The degree bound is automatic on the square lattice; it imposes no additional restriction on the selected edges. Write $n = |V|$ and $m = |F|$. At v put the square $C_v = 3v + [-1/2, 1/2]^2$. For the edge from v to $v + (1, 0)$ put

$$R_e = 3v + [1/2, 5/2] \times [-1/4, 1/4],$$

and use the rotated rectangle for a vertical edge. Define

$$\Omega(V,F) = \operatorname{int}\left(\bigcup_{v\in V} C_v \;\cup\; \bigcup_{e\in F} R_e\right). \tag{2.2}$$

The pieces have disjoint interiors. Distinct passages can meet only through their listed endpoint chambers: a horizontal passage avoids the vertical lattice lines by half a unit at both ends, and conversely for a vertical passage. The windows lie strictly inside chamber faces. Hence (2.2) is a bounded polygonal open set with Lipschitz boundary, possibly disconnected, and

$$|\Omega(V,F)| = |V| + |F|. \tag{2.3}$$

For every such choice the conclusion of this paper is

$$\boxed{\begin{aligned} N_{\Omega(V,F)}(E) &\le \frac{(|V|+|F|)E}{4\pi} \quad (E>0),\\ \lambda_k(\Omega(V,F)) &\ge \frac{4\pi k}{|V|+|F|} \quad (k\ge 1). \end{aligned}} \tag{2.4}$$

Here and throughout, $N_D(E) = \#\{j : \lambda_j(D) \le E\}$ includes multiplicities and equality at the endpoint. The two assertions are equivalent: apply the eigenvalue assertion to $j = N_D(E) > 0$ in one direction (the zero-count case is immediate), and the counting assertion at $E = \lambda_k(D)$ in the other. The proof of (2.4) is completed by the cell estimates below. Neither connectivity nor a positive proportion of selected edges is needed.

2.2. **A compact family with many holes.** Take $V_q = \{0,\dots,q-1\}^2$, $q \ge 2$, and include every nearest-neighbour edge. Write Ω_q for the resulting connected domain. The room and passage counts, area, and hole count are

$$n_q = q^2, \qquad m_q = 2q(q-1), \tag{2.5}$$

$$A_q = 3q^2 - 2q, \qquad h_q = (q-1)^2 = m_q - n_q + 1. \tag{2.6}$$

Indeed the bounded faces of the embedded grid are precisely its $(q-1)^2$ elementary squares. Thickening the vertices and edges as in (2.2) leaves these distinct complementary components and creates no extra contact.

Each hole has area 6. For the face with chamber centers $(0,0),(3,0),(3,3),(0,3)$, the square between the inner passage sides has side $5/2$. Each corner room occupies a further square of side $1/4$ inside it. The remaining area is $25/4 - 4/16 = 6$. The four extreme chamber corners show that the convex hull of $\overline{\Omega_q}$ is the square of side $3q-2$. Consequently

$$\operatorname{diam}(A_q^{-1/2}\Omega_q) \le \frac{\sqrt{2}(3q-2)}{\sqrt{3q^2-2q}} < \sqrt{6}. \tag{2.7}$$

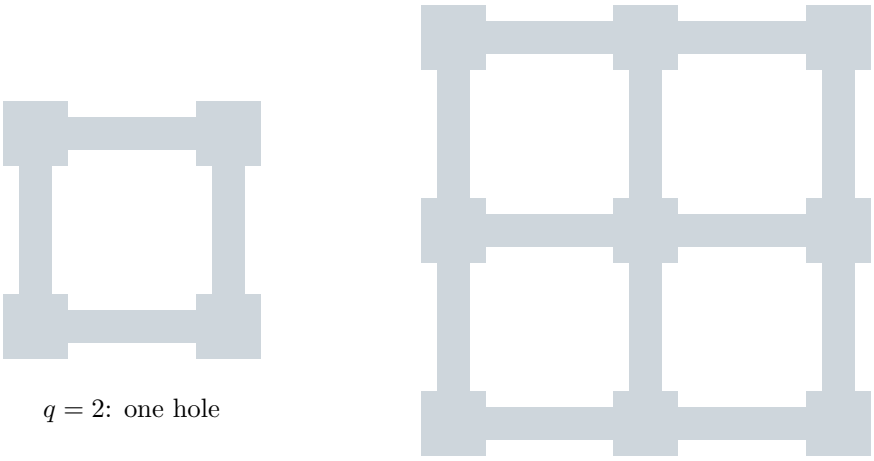
The strict inequality follows from $6A_q - 2(3q-2)^2 = 12q - 8 > 0$.

The boundary length nevertheless grows after area normalization. Each room contributes perimeter four, and each passage contributes five before its two interfaces are removed twice from the sum. Since each interface has length $1/2$, the physical perimeter is

$$|\partial\Omega_q| = 4n_q + 3m_q = 10q^2 - 6q, \qquad |\partial(A_q^{-1/2}\Omega_q)| = \frac{10q^2-6q}{\sqrt{3q^2-2q}} \longrightarrow \infty.$$

Thus the same local certificate applies despite increasing boundary length and hole count within a bounded area-normalized diameter.

These domains do not tile the plane by congruent copies. Another connected open copy with disjoint interior cannot contain a boundary point of Ω_q, since its openness would force intersection with the interior of Ω_q. A copy meeting a bounded hole must therefore lie entirely inside that hole. This is impossible because $A_q \ge 8 > 6$. Tile boundaries cannot fill the open hole: the tiling is locally finite by packing a fixed interior disk from each bounded copy in a



$q = 2$: one hole

$q = 3$: four holes

FIGURE 1. The actual domains Ω_2 and Ω_3, drawn at the same physical scale. Shading is the membrane, not a center-line graph. All white regions are outside it. Each room has side one; each passage has length two and width 1/2. The middle chamber in the right-hand domain is a genuine four-port junction.

bounded enlargement, and polygon boundaries have area zero. Composing a finite congruent partition with a plane tiling of its parent would produce the forbidden tiling as well.

There is also a useful quantitative distinction from square-removal criteria. Every containing rectangle T, in any orientation, contains the convex hull, and therefore

$$\frac{A_q}{|T|} \le \frac{3q^2 - 2q}{(3q-2)^2} \le \frac{1}{2}. \tag{2.8}$$

The last inequality is $(q-2)(3q-2) \ge 0$. A stronger comparison allows connected nonrectangular tiling parents. Let P be a bounded connected Lipschitz plane tile containing Ω_q. In a tiling with one distinguished copy P, every other copy is disjoint from $\overline{\Omega_q}$. A copy entering a bounded hole must stay inside it, again contradicting $|P| \ge A_q > 6$. Thus P fills almost all of every hole and

$$|P| \ge A_q + 6h_q, \qquad \frac{|P| - A_q}{|P|} \ge \frac{6h_q}{A_q + 6h_q} \ge \frac{3}{7}. \tag{2.9}$$

Here $8h_q - A_q = (q-2)(5q-4) \ge 0$.

Suppose P is a congruent M-subtile of a strip tile T, and deleting finitely many interior-disjoint squares of side lengths ℓ_i from P gives Ω_q. Let w be the strip width and S its side-face length, so $Sw = M|P|$. A square of side ℓ_i has transverse width at least ℓ_i, even after rotation, hence $\ell_i \le w$. The planar sufficient condition

$$S \ge 2\sqrt{2}\pi M \sum_i \ell_i$$

in [5, Theorem 1.15 and Corollary 1.16] (numbering of arXiv version 4) would imply

$$|P| - A_q = \sum_i \ell_i^2 \le w \sum_i \ell_i \le \frac{|P|}{2\sqrt{2}\pi},$$

contrary to (2.9). The case $M = 1$ uses the theorem directly. This comparison concerns that stated finite-square sufficient criterion with connected parents; it is not a classification of all possible spectral inheritance constructions.

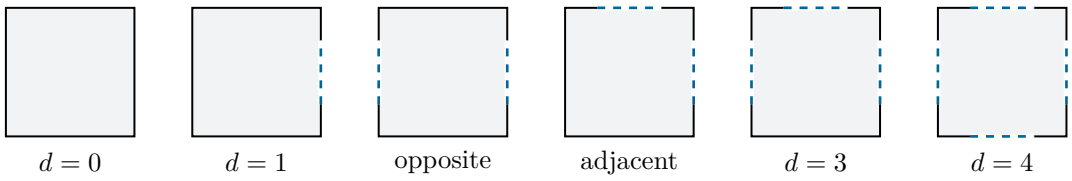


FIGURE 2. The six scaled chamber problems in (2.10). Each square has side two. Solid black segments are Dirichlet; dashed blue segments are Neumann windows of length one. The three-port cell has its absent port below, as in the cluster calculation; the final cell has four Neumann windows.

2.3. **Forms, cuts, and the six chamber types.** For a bounded Lipschitz cell D with a relatively open Dirichlet part Γ_D, the mixed form is

$$a_D(u) = \int_D |\nabla u|^2 \, \mathrm{d}x, \qquad \mathcal{V}_D = \{u \in H^1(D) : \operatorname{Tr} u = 0 \text{ on } \Gamma_D\}.$$

The remaining boundary has the natural Neumann condition. Changing the status of finitely many corner points does not change the trace condition. The form has compact resolvent, and its eigenvalues are understood variationally. All cells below retain Dirichlet boundary of positive length.

Dilate the network by two. Let $C = (-1, 1)^2$, and let S specify a subset of its four faces. For each face in S, its centered segment of length one is Neumann; every other boundary segment is Dirichlet. Denote this mixed problem by C_S, and put $d = |S|$. Let $R = (0, 4) \times (0, 1)$ have Neumann conditions at its ends and Dirichlet conditions on its long sides. The local assertion is

$$N_{C_S}(E) + \frac{d}{2} N_R(E) \le \frac{(4 + 2d)E}{4\pi}, \qquad E > 0, \quad d \le 4. \tag{2.10}$$

There are six types up to the symmetries of the square, since two adjacent and two opposite ports are different.

To see why (2.10) proves the physical result, restrict a function in $H^1_0(2\Omega)$ to every chamber and passage. Each restriction belongs to the corresponding mixed form domain. Norms and energies add exactly, while the direct sum allows independent traces at the cut interfaces. The min–max principle gives

$$N_{2\Omega}(E) \le \sum_{v \in V} N_{C_{S_v}}(E) + m N_R(E).$$

Since $\sum_v d(v) = 2m$, summing (2.10) bounds the right side by $(4n + 4m)E/(4\pi)$. Finally, $\lambda_k(2\Omega) = \lambda_k(\Omega)/4$, so this is exactly (2.4). In particular, an absent edge has neither a Neumann window in C_{S_v} nor an assigned passage share. The proof does not rely on boundary vertices being negligible compared with interior vertices.

2.4. **The separable comparison.** Enlarge each Neumann window of C_S to its entire face; write H_S for the resulting rectangular mixed problem. Then $N_{C_S}(E) \le N_{H_S}(E)$. On an interval of length two, the frequencies divided by π are $(j + \theta)/2$, $j \ge 0$, with

$$\theta = 0 \quad (\mathrm{NN}), \qquad \theta = \tfrac{1}{2} \quad (\mathrm{DN}), \qquad \theta = 1 \quad (\mathrm{DD}).$$

Thus $\theta_x + \theta_y = 2 - d/2$. Put $E = \pi^2 r^2$ and $P = 16r^2$. The chamber events are $P = a^2 + b^2$, where the allowed nonnegative numerators are

Type	a	b	d
No port	$2,4,6,\dots$	$2,4,6,\dots$	0
One port	$2,4,6,\dots$	$1,3,5,\dots$	1
Opposite ports	$2,4,6,\dots$	$0,2,4,\dots$	2
Adjacent ports	$1,3,5,\dots$	$1,3,5,\dots$	2
Three ports	$1,3,5,\dots$	$0,2,4,\dots$	3
Four ports	$0,2,4,\dots$	$0,2,4,\dots$	4

Interchanging coordinates does not affect the counts. For R, the exact levels are $\pi^2(m^2+n^2/16)$ with $m \geq 1$, $n \geq 0$; its events are $P = 16m^2 + n^2$. The zero longitudinal mode is included.

The remainder of the proof separates two tasks. First, actual Dirichlet segments give quantitative lower bounds for the few levels that need improvement. Second, exact counting and an analytic tail show that these improvements suffice at every energy.

3. The boundary estimates at the exceptional levels

We work throughout this section at the dilated scale: the chamber is $C = (-1,1)^2$, and every present port is a centred interval of length one. For a set S of present ports, let C_S denote the mixed chamber with Neumann condition on these intervals and Dirichlet condition on the rest of its boundary. Let H_S denote the comparison chamber obtained by extending each present Neumann interval to its entire face. The quadratic form is the Dirichlet integral on the corresponding closed subspace of $H^1(C)$. Consequently, increasing the Dirichlet part increases every eigenvalue, with multiplicity. All counting functions below include equality at the energy in question.

Only three estimates beyond this elementary comparison will be needed. The first is a common lower bound for the ground state. The second moves one state of the chamber with two opposite ports. The third moves both members of a double level of the chamber with three ports. We give the boundary calculations explicitly, including the entire spectral complement in the last estimate.

3.1. **A common ground-state estimate.** Let C_4 be the chamber with all four Neumann windows. Every C_S is a Dirichlet strengthening of C_4. It therefore suffices to prove

$$\lambda_1(C_4) \geq \frac{2\pi^2}{\pi^2+2} > \frac{\pi^2}{6}. \tag{3.1}$$

The last strict inequality follows from $\pi^2 < 10$.

Consider the unit quadrant $Q = (0,1)^2$. Its prescribed Dirichlet set is

$$D_Q = (\{1\} \times [1/2,1]) \cup ([1/2,1] \times \{1\}).$$

No boundary condition on the two coordinate axes is required in the following calculation. Define

$$f(t) = \mathbf{1}_{(1/2,1)}(t), \qquad H(t) = \frac{1}{2}\min(t, 1-t), \qquad F(x,y) = \big(xf(y) + H(x),\ yf(x) + H(y)\big).$$

Since $H' = 1/2 - f$ almost everywhere, $\operatorname{div} F = 1$ in each of the four half-size squares. Across each internal segment, the normal component of F is continuous: its jump is tangential. Thus the same divergence identity holds distributionally on Q. The outward normal component of F is one on D_Q and zero on the remaining outer boundary.

The norm of this vector field is exact. Indeed,

$$\int_0^1 H(t)^2\,dt = \frac{1}{48}, \qquad \int_0^1 tH(t)\,dt = \frac{1}{16}, \qquad \int_0^1 f(t)\,dt = \int_0^1 f(t)^2\,dt = \frac{1}{2}.$$

It follows that

$$\int_Q |F|^2 = 2\left(\frac{1}{6} + \frac{1}{16} + \frac{1}{48}\right) = \frac{1}{2}. \tag{3.2}$$

For $u \in H^1(Q)$ whose trace vanishes on D_Q, integration by parts on the four half-size squares, with cancellation of the internal normal fluxes, gives

$$\int_Q u = -\int_Q F \cdot \nabla u, \qquad \left|\int_Q u\right|^2 \le \frac{1}{2}\int_Q |\nabla u|^2.$$

These identities extend from smooth restrictions to $H^1(Q)$ by the trace theorem and density. As $|Q| = 1$, the mean-zero Neumann Poincaré inequality on the unit square yields

$$\int_Q \left|u - \int_Q u\right|^2 \le \frac{1}{\pi^2}\int_Q |\nabla u|^2.$$

For completeness, its constant follows by expanding the mean-zero function in the cosine basis: every nonconstant product mode has eigenvalue at least π^2. The orthogonal decomposition into the mean and the mean-zero part now gives

$$\int_Q |u|^2 \le \left(\frac{1}{2} + \frac{1}{\pi^2}\right)\int_Q |\nabla u|^2.$$

Every admissible function on C_4 restricts to four functions of this kind after reflecting their quadrants onto Q. Summing the four inequalities proves (3.1). In particular,

$$N^{\le}_{\widetilde{C}_S}(E) = 0 \qquad \left(0 < E \le \frac{\pi^2}{6}\right) \tag{3.3}$$

for every set of present ports.

3.2. **An anchored trace estimate and the opposite-port chamber.** We first record the variational calculation used for the simple level. Suppose a comparison problem on a rectangle has Dirichlet condition at $x = 0$. Let ϕ be a normalized simple eigenfunction at its kth eigenvalue λ_0, and suppose the next eigenvalue is $\lambda_0 + \Delta$, where $\Delta > 0$. Add a Dirichlet condition on a measurable portion Γ of the opposite face $x = L$, and write

$$g = \int_\Gamma |\phi|^2 > 0.$$

Then the strengthened problem satisfies

$$\lambda_k^{\text{new}} \ge \lambda_0 + \frac{\Delta g}{g + 2\sqrt{\lambda_0 + \Delta}}. \tag{3.4}$$

To see this, impose orthogonality to the first $k-1$ comparison eigenfunctions on the strengthened form domain and write

$$u = a\phi + h, \qquad q(h) = \int |\nabla h|^2 - \lambda_0 \int |h|^2.$$

Here h is orthogonal to every comparison mode at or below λ_0, so

$$q(u) = q(h), \qquad \|h\|_2^2 \le \frac{q(h)}{\Delta}, \qquad \|\nabla h\|_2^2 \le \frac{\lambda_0 + \Delta}{\Delta} q(h).$$

The identity for $q(u)$ follows from the weak eigenfunction equation for ϕ. Since h vanishes at $x = 0$, the fundamental theorem of calculus on almost every horizontal slice gives, for every $\tau > 0$,

$$\int_{x=L} |h|^2 = 2\,\mathrm{Re}\int \overline{h}\,\partial_x h \le \tau\|\partial_x h\|_2^2 + \tau^{-1}\|h\|_2^2.$$

On Γ we have $h = -a\phi$. Consequently,

$$|a|^2 g \le \frac{\tau(\lambda_0 + \Delta) + \tau^{-1}}{\Delta}\, q(h), \qquad \|u\|_2^2 \le \frac{g + \tau(\lambda_0 + \Delta) + \tau^{-1}}{\Delta g}\, q(h).$$

Taking $\tau = (\lambda_0 + \Delta)^{-1/2}$ and applying the max–min principle proves (3.4). The lower comparison eigenfunctions need not belong to the strengthened form domain. They serve only as L^2 vectors defining the $k-1$ orthogonality constraints, which is precisely what the max–min principle permits.

Now place the two opposite ports on the faces $x = \pm 1$. Both the actual chamber and its comparison problem are invariant under reflection in each coordinate axis. Their four parity sectors are orthogonal reducing subspaces. In the sector odd in both variables, restriction to $Q = (0,1)^2$ gives Dirichlet conditions at $x = 0$ and $y = 0, 1$. The comparison condition at $x = 1$ is Neumann; the actual chamber adds Dirichlet condition on

$$\Gamma = \{1\} \times [1/2, 1].$$

The comparison eigenvalues in this sector are

$$\pi^2((m + 1/2)^2 + n^2), \qquad m \geq 0, \quad n \geq 1.$$

Its normalized ground state, ground eigenvalue, and next gap are

$$\phi(x, y) = 2\sin(\pi x/2)\sin(\pi y), \qquad \lambda_0 = \frac{5\pi^2}{4}, \qquad \Delta = 2\pi^2.$$

The normalization is on the unit quadrant, so the trace mass is

$$g = 4\int_{1/2}^{1} \sin^2(\pi y)\, dy = 1.$$

Formula (3.4) therefore gives

$$\lambda_1^{--}(C_S) \geq \frac{5\pi^2}{4} + \frac{2\pi^2}{1 + \pi\sqrt{13}} > \frac{41\pi^2}{32}. \tag{3.5}$$

For the last inequality, $\pi < 22/7$ and $\sqrt{13} < 4$ give $1 + \pi\sqrt{13} < 95/7 < 64$.

All other sector eigenvalues retain their comparison lower bounds. The comparison odd–odd sector has exactly one eigenvalue between $5\pi^2/4$ and $41\pi^2/32$, while the actual sector has none. Accordingly,

$$N_{\widetilde{C}_S}^{\leq}(E) \leq N_{\widetilde{H}_S}^{\leq}(E) - 1 \qquad \left(\frac{5\pi^2}{4} \leq E \leq \frac{41\pi^2}{32}\right) \tag{3.6}$$

when the two ports are opposite.

3.3. **The double level of the three-port chamber.** Place the missing port on $y = -1$, so this entire face is Dirichlet. Reflection in $x = 0$ splits the actual and comparison chambers into even and odd sectors. In the even sector, put $t = y + 1$ and work on $Q = (0, 1) \times (0, 2)$. The comparison problem has Neumann conditions at $x = 0, 1$ and $t = 2$, and Dirichlet condition at $t = 0$. An orthonormal basis is

$$\psi_{mn}(x, t) = c_m \cos(m\pi x) \sin\frac{(2n + 1)\pi t}{4}, \quad c_0 = 1, \quad c_m = \sqrt{2}\ (m \geq 1), \tag{3.7}$$

where $m, n \geq 0$, with eigenvalues

$$\lambda_{mn} = \pi^2\left(m^2 + \frac{(2n + 1)^2}{16}\right).$$

The three eigenvalues below $25\pi^2/16$ are $\pi^2/16$, $9\pi^2/16$, and $17\pi^2/16$. The level $\lambda_* = 25\pi^2/16$ is double, with normalized eigenfunctions

$$\phi_A(x, t) = \sin(5\pi t/4), \qquad \phi_B(x, t) = \sqrt{2}\cos(\pi x)\sin(3\pi t/4).$$

The next comparison level is $41\pi^2/16$, so the gap above the double level is π^2.

The actual added Dirichlet boundary contains

$$\Gamma = \{1\} \times J, \qquad J = [0, 1/2] \cup [3/2, 2].$$

We use only this part of the physical Dirichlet boundary; allowing arbitrary traces on the additional Dirichlet portions gives a valid lower comparison. Impose orthogonality to the first three comparison even modes, and decompose any admissible function as

$$\begin{aligned} u &= a\phi_A + b\phi_B + h, \\ q(h) &= \int_Q |\nabla h|^2 - \lambda_* \int_Q |h|^2 \geq \pi^2 \|h\|_2^2. \end{aligned} \tag{3.8}$$

Thus h is orthogonal to all five comparison modes at or below λ_*. As before, $q(u) = q(h)$. The boundary moments. Define the two linear functionals

$$L_A(v) = \int_J \phi_A(1,t)v(1,t)\,dt, \qquad L_B(v) = \int_J \phi_B(1,t)v(1,t)\,dt.$$

All eigenfunctions appearing here are real; these expressions are linear also for complex v. For $a, n \geq 0$, set

$$s_a(t) = \sin\frac{(2a+1)\pi t}{4}, \qquad I_{an} = \int_J s_a(t)s_n(t)\,dt.$$

The substitution $t \mapsto 2-t$ gives $s_a(2-t) = (-1)^a\cos((2a+1)\pi t/4)$. Therefore

$$I_{an} = \begin{cases} \displaystyle\int_0^{1/2} \cos\frac{(a-n)\pi t}{2}\,dt, & a+n \text{ even}, \\ \displaystyle -\int_0^{1/2} \cos\frac{(a+n+1)\pi t}{2}\,dt, & a+n \text{ odd}. \end{cases} \tag{3.9}$$

In particular,

$$I_{11} = I_{22} = \frac{1}{2}, \qquad I_{20} = \frac{1}{\pi}, \qquad I_{10} = -\frac{1}{\pi}, \qquad I_{12} = I_{21} = 0.$$

Since $\phi_B(1,t) = -\sqrt{2}\,s_1(t)$, the matrix of these functionals on the double eigenspace is exactly

$$\begin{pmatrix} L_A(\phi_A) & L_A(\phi_B) \\ L_B(\phi_A) & L_B(\phi_B) \end{pmatrix} = D := \begin{pmatrix} 1/2 & 0 \\ 0 & 1 \end{pmatrix}. \tag{3.10}$$

The trace condition on u consequently implies

$$L(h) := \begin{pmatrix} L_A(h) \\ L_B(h) \end{pmatrix} = -D\begin{pmatrix} a \\ b \end{pmatrix}. \tag{3.11}$$

The functions s_n, $n \geq 0$, form an orthonormal basis of $L^2(0,2)$. Applying Parseval to $\mathbf{1}_J s_a$ gives

$$\sum_{n\geq 0} I_{an}^2 = \int_J s_a^2 = \frac{1}{2}.$$

Subtracting the three explicitly computed terms yields the exact complement identity

$$\sum_{n\geq 3} I_{an}^2 = \frac{1}{4} - \frac{1}{\pi^2}, \qquad a = 1, 2. \tag{3.12}$$

The complete high-mode bound. Let $\mathcal{I}$ consist of the pairs (m,n) with $\lambda_{mn} > \lambda_*$. The positive boundary Gram matrix is

$$G = \sum_{(m,n)\in\mathcal{I}} \frac{L(\psi_{mn})L(\psi_{mn})^*}{\lambda_{mn} - \lambda_*}. \tag{3.13}$$

Here the star denotes conjugate transpose. The diagonal estimates below prove convergence of this full series. Weighted Cauchy–Schwarz in the expansion of h then gives

$$|z^*L(h)|^2 \leq (z^*Gz)\,q(h) \qquad (z \in \mathbb{C}^2). \tag{3.14}$$

One may first apply this inequality to finite expansions. Their spectral partial sums converge in the form norm, and the trace functionals are continuous in that norm, so the displayed inequality follows for every h in (3.8).

At $x = 1$ the boundary values of ψ_{mn} are $c_m(-1)^m s_n$. Hence

$$G_{AA} = \frac{1}{\pi^2}\sum_{n\geq 0} I_{2n}^2 H_n, \qquad G_{BB} = \frac{2}{\pi^2}\sum_{n\geq 0} I_{1n}^2 H_n, \tag{3.15}$$

where the following expressions remove precisely the five low modes:

$$\begin{aligned} H_0 &= \sum_{m\ge 2} \frac{2}{m^2-3/2}, \qquad H_1 = \sum_{m\ge 2} \frac{2}{m^2-1}, \qquad H_2 = \sum_{m\ge 1} \frac{2}{m^2}, \\ H_n &= \frac{1}{\beta_n} + \sum_{m\ge 1} \frac{2}{m^2+\beta_n} \quad (n\ge 3), \qquad \beta_n = \frac{(2n+1)^2-25}{16} \ge \frac{3}{2}. \end{aligned} \tag{3.16}$$

The first three are bounded or evaluated by

$$H_0 \le \frac{16}{5}\left(\frac{\pi^2}{6}-1\right) < \frac{32}{15}, \qquad H_1 = \frac{3}{2}, \qquad H_2 = \frac{\pi^2}{3}. \tag{3.17}$$

For H_0 we used $m^2-3/2 \ge (5/8)m^2$ for $m\ge 2$. For H_1 the identity $2/(m^2-1) = 1/(m-1) - 1/(m+1)$ telescopes. For $n\ge 3$, comparison of a decreasing positive summand with its integral gives

$$H_n \le \frac{1}{\beta_n} + 2\int_0^\infty \frac{dx}{x^2+\beta_n} = \frac{1}{\beta_n} + \frac{\pi}{\sqrt{\beta_n}} \le \frac{2}{3} + \pi\sqrt{\frac{2}{3}} < \frac{23}{7}. \tag{3.18}$$

The last comparison uses $\sqrt{2/3} < 5/6$ and $\pi < 22/7$.

Put

$$A = \frac{H_0}{\pi^4} + \frac{23}{7\pi^2}\left(\frac{1}{4} - \frac{1}{\pi^2}\right).$$

The bounds $\pi > 25/8$ and $\pi^2 < 10$ give

$$A < \left(\frac{64}{625}\right)^2 \frac{32}{15} + \frac{64}{625}\frac{23}{7}\frac{3}{20} = \frac{2987504}{41015625} < \frac{73}{1000}. \tag{3.19}$$

Using (3.12) in (3.15), we obtain

$$G_{AA} < \frac{1}{12} + \frac{73}{1000} = \frac{469}{3000}, \qquad G_{BB} < 2\left(\frac{73}{1000} + \frac{3}{8\pi^2}\right) < \frac{557}{2500}. \tag{3.20}$$

No off-diagonal estimate is needed. The largest eigenvalue of the positive matrix $D^{-1}GD^{-1}$ is at most its trace, and

$$\operatorname{Tr}(D^{-1}GD^{-1}) = 4G_{AA} + G_{BB} < \frac{6361}{7500} < \frac{17}{20}.$$

Congruence by D gives the uniform matrix bound

$$G \le \frac{17}{20} D^2. \tag{3.21}$$

Every term of the infinite spectral complement has entered these estimates. In particular, no truncation of a positive tail is being used as an upper bound.

The spectral conclusion, including multiplicity. Let $v = (a,b)^{\mathsf{T}}$. If $v \ne 0$, use $z = D^{-1}v$ in (3.14), together with (3.11) and (3.21). This gives

$$|v|^4 \le \frac{17}{20}|v|^2 q(h), \qquad |a|^2 + |b|^2 \le \frac{17}{20} q(h).$$

The second inequality is also valid when $v = 0$. Combining it with the gap in (3.8), we find

$$\|u\|_2^2 = |a|^2 + |b|^2 + \|h\|_2^2 \le \left(\frac{17}{20} + \frac{1}{\pi^2}\right) q(h).$$

The three orthogonality conditions imposed above are legitimate in the max–min formula for the fourth actual even eigenvalue. We conclude that

$$\begin{aligned} \lambda_4^+(C_S) &\ge \frac{25\pi^2}{16} + \frac{20\pi^2}{17\pi^2+20} \\ &> \frac{25\pi^2}{16} + \frac{2\pi^2}{19} = \frac{507\pi^2}{304} > \frac{5\pi^2}{3} > \frac{53\pi^2}{32}. \end{aligned} \tag{3.22}$$

Here $\pi^2 < 10$, $507/304 - 5/3 = 1/912$, and $5/3 - 53/32 = 1/96$ prove the strict comparisons. This is a uniform estimate for the Rayleigh quotient on the constrained space. Both members of the double level, and every later even eigenvalue, therefore lie strictly above $53\pi^2/32$.

For the odd sector, the comparison eigenvalues are

$$\pi^2\left((m+1/2)^2+\frac{(2n+1)^2}{16}\right), \qquad m,n\geq 0.$$

Exactly two lie at or below $53\pi^2/32$, namely $5\pi^2/16$ and $13\pi^2/16$; the next is $29\pi^2/16$. Thus the actual even sector contributes at most three states and the odd sector at most two through $53\pi^2/32$. On the interval starting at $25\pi^2/16$, the full comparison chamber has exactly seven states. Consequently,

$$N^{\leq}_{\widetilde{C}_S}(E)\leq 5 = N^{\leq}_{\widetilde{H}_S}(E)-2 \qquad \left(\frac{25\pi^2}{16}\leq E\leq \frac{53\pi^2}{32}\right) \tag{3.23}$$

for the three-port chamber.

In the event coordinate $P = 16E/\pi^2$, the preceding estimates allow one to remove any comparison ground state below $P = 8/3$, one state of the opposite-port chamber for $20 \leq P < 41/2$, and two states of the three-port chamber for $25 \leq P < 53/2$. The estimates remain valid at the upper endpoints. The finite count may nevertheless cease using a correction there, provided the resulting jump of the chosen upper bound is checked at that endpoint.

4. An elementary estimate for the counting-function tail

The boundary estimates handle only a few low levels. To control every remaining energy, we first obtain a uniform lower bound on the fractional parts lost when a quarter-circle sum is replaced by its integral. The uniformity in both sampling phases is useful here: the chamber types have different Dirichlet and Neumann conditions, but they can all be treated by the same estimate.

4.1. **Fractional parts in consecutive blocks.** Write $\{z\} = z - \lfloor z\rfloor$. For an integer $K \geq 2$, set

$$M_K(\alpha)=\min_{t\in\mathbb{R}}\sum_{k=0}^{K-1}\{t-k\alpha\}, \qquad C_K=\frac{K(K-1)}{2}.$$

As t runs through a period, the sum increases linearly between its downward jumps. Its minimum therefore occurs when at least one fractional part is zero, including when several jumps coincide. Consequently

$$M_K(\alpha)=\min_{0\leq j<K}\sum_{k=0}^{K-1}\{(j-k)\alpha\}, \qquad 0\leq M_K(\alpha)\leq\frac{K-1}{2}. \tag{4.1}$$

For the upper bound, average the K candidate sums. Each unordered pair of distinct indices contributes $\{z\}+\{-z\}$, which is either zero or one; their total is at most C_K.

Although a fractional part is discontinuous, this minimized sum is C_K-Lipschitz. Indeed, if $\beta \geq \alpha$, choose a minimizing phase for β and use

$$\{z+h\}\leq\{z\}+h \qquad (h\geq 0)$$

with $h = k(\beta - \alpha)$. This gives $M_K(\alpha) \leq M_K(\beta) + C_K(\beta - \alpha)$. Reversing the order of the K terms and translating the phase shows $M_K(-\alpha) = M_K(\alpha)$; applying the same inequality to $-\beta \leq -\alpha$ gives the reverse bound. Thus

$$|M_K(\beta)-M_K(\alpha)|\leq C_K|\beta-\alpha|. \tag{4.2}$$

This argument includes rational slopes and repeated phases.

For $T > 0$ and $a > 0$, define

$$f_T(t)=\sqrt{T^2-t^2}, \qquad g_a(x)=\frac{ax}{\sqrt{1-x^2}}, \qquad x_G=\frac{G}{\sqrt{a^2+G^2}},$$

where $G > 0$, and put

$$A(a, K, G) = \frac{1}{K} \int_0^{x_G} M_K(g_a(x))\, dx.$$

For every $\theta \in [0, 1]$ and every $\eta \in \mathbb{R}$, the complete sampled fractional-part sum satisfies

$$\sum_{\substack{j \geq 0 \\ j+\theta \leq T}} \{a f_T(j + \theta) - \eta\} \geq TA(a, K, G) - \frac{K^2 G}{8} - \frac{K}{2}. \tag{4.3}$$

Write $h(t) = a f_T(t)$ and first consider a block $I = [b, b + K]$ contained in $[0, Tx_G]$, with samples $s_k = b + k + \frac{1}{2}$, $0 \leq k < K$. For any $t \in I$, the tangent line to the concave function h gives

$$h(s_k) = h(t) + h'(t)(s_k - t) - e_k(t), \qquad e_k(t) \geq 0.$$

Its values at successive samples form an arithmetic progression with drop $g_a(t/T)$. Since $\{z - e\} \geq \{z\} - e$ for $e \geq 0$,

$$\sum_{k=0}^{K-1} \{h(s_k) - \eta\} \geq M_K(g_a(t/T)) - \sum_{k=0}^{K-1} e_k(t).$$

Average over $t \in I$. Put $q(s) = -h''(s) \geq 0$, write $x = s - b$, and let n be the number of sample points at or to the left of s. Changing the order of integration in the nonnegative tangent remainders gives

$$\frac{1}{K} \int_I \sum_k e_k(t)\, dt = \int_I w_K(s - b) q(s)\, ds,$$

where

$$\begin{aligned} w_K(x) &= \frac{x}{K} \sum_{s_k \geq s} (s_k - s) + \frac{K - x}{K} \sum_{s_k \leq s} (s - s_k) \\ &= \frac{x(K - x)}{2} - \frac{(n - x)^2}{2} \leq \frac{K^2}{8}. \end{aligned} \tag{4.4}$$

Indeed, the left and right distance sums are $nx - n^2/2$ and $K^2/2 - Kx + nx - n^2/2$. The formula also holds at a sampled point, where the corresponding summand is zero. Therefore the complete block obeys

$$\begin{aligned} \sum_k \{h(s_k) - \eta\} \geq \frac{1}{K} \int_I M_K(g_a(t/T))\, dt \\ - \frac{K^2}{8} [g_a((b + K)/T) - g_a(b/T)]. \end{aligned}$$

For the prescribed sample phase, take $j_0 = 0$ when $\theta \geq 1/2$ and $j_0 = 1$ otherwise. Then $b = j_0 + \theta - 1/2 \in [0, 1)$, so consecutive blocks starting at b are centered on original samples. Retain all complete blocks ending at or before Tx_G. Their derivative increments telescope to at most G. The omitted initial and final lengths total less than $K + 1$. By (4.1), the corresponding loss from the integral divided by K is less than $(K + 1)(K - 1)/(2K) < K/2$. All discarded fractional parts are nonnegative. If there is no complete block, $Tx_G < K + 1$ and the same bound gives $TA < K/2$ directly. This proves (4.3) for all phases, small T, and empty sample sets. All differentiations occur below $Tx_G < T$, where the circle is smooth.

For degrees at most three, use

$$A(1, 6, 2) \geq \frac{860001557153}{3276800000000} > \frac{13}{50}, \tag{4.5}$$

$$A(4, 4, 4) \geq \frac{138748241337}{819200000000} > \frac{169}{1000}. \tag{4.6}$$

These are exact lower sums certified by the Lipschitz bound. The integer construction and its complete parameters are given in Appendix A. Inserting them into (4.3) gives, respectively,

$$F_H \geq \frac{13}{25} r - 12, \qquad F_R \geq \frac{169}{1000} r - 10, \tag{4.7}$$

for chamber rows with $T = 2r$, $a = 1$, and corridor rows with $T = r$, $a = 4$.

4.2. **Shifted quarter-circle sums and the common tail.** Let $c = \sqrt{2/27}$. The following elementary estimate applies to $\theta = 0, \frac{1}{2}, 1$:

$$\sum_{\substack{j\geq 0\\ j+\theta\leq T}} f_T(j+\theta) \leq \frac{\pi T^2}{4} + \left(\frac{1}{2} - \theta\right) T + c\sqrt{T}. \tag{4.8}$$

For $\theta = 1$, write $M = \lfloor T \rfloor$ and $s = T - M$. The trapezoidal inequality for the concave function f_T, followed by the chord bound on $[M, T]$, gives

$$\begin{aligned}\sum_{m=1}^{M} f_T(m) &\leq \frac{\pi T^2}{4} - \frac{T}{2} + \frac{f_T(M)}{2} - \int_M^T f_T(t)\,dt \\ &\leq \frac{\pi T^2}{4} - \frac{T}{2} + \frac{1-s}{2}\sqrt{2Ts}.\end{aligned}$$

The last factor is at most $c\sqrt{T}$, since $\max_{0\leq s\leq 1}(1-s)\sqrt{s} = 2/(3\sqrt{3})$. When $M = 0$, the initial trapezoidal identity has both sides zero before the tail is inserted, so the argument still applies. The case $\theta = 0$ adds $f_T(0) = T$. For $\theta = \frac{1}{2}$, start the trapezoids at $1/2$. The initial correction is nonpositive because $\int_0^{1/2} f_T(t)\,dt \geq f_T(1/2)/2$; the last correction is the same expression with s the distance from the final sampled point to T. If there is no half-integer sample, the right-hand side is positive and the assertion is immediate.

Recall the dilated chamber C_S, with $d = |S|$ present ports, and the 4-by-1 corridor R. Relax each present-port face of C_S entirely to Neumann, keeping every missing-port face entirely Dirichlet, and call the resulting rectangle H_S. In one coordinate its frequencies, in units of π, are $(j+\theta)/2$, $j \geq 0$, where

$$\theta = 0,\ \tfrac{1}{2},\ 1 \quad \text{for NN, DN, DD, respectively.}$$

The two coordinate phases satisfy $\theta_x + \theta_y = 2 - d/2$. Throughout this calculation

$$E = \pi^2 r^2, \qquad P = 16r^2, \tag{4.9}$$

and all counts include equality at the threshold.

With $T = 2r$, exact counting in each horizontal row yields

$$N_{H_S}^{\leq}(E) = \sum_{\substack{j\geq 0\\ j+\theta_y\leq T}} \left[f_T(j+\theta_y) + 1 - \theta_x - \{f_T(j+\theta_y) - \theta_x\}\right].$$

The summand is $\lfloor f_T(j+\theta_y) - \theta_x \rfloor + 1$. Since $0 \leq \theta_x \leq 1$ and $f_T \geq 0$, it is never negative and equals the exact number of allowed nonnegative indices. In particular, a row below the first DN frequency contributes zero; a zero-radius NN row contributes one; a zero-radius DD row contributes zero. There is no endpoint convention to repair afterward.

The number of rows is at most $T + 1 - \theta_y$. Multiplying this bound by $1 - \theta_x \geq 0$ and using (4.8), with F_H the fractional-part sum just displayed, gives

$$N_{H_S}^{\leq}(E) \leq \pi r^2 + (d-1)r + c\sqrt{2r} + 1 - F_H. \tag{4.10}$$

Here the constant $(1-\theta_x)(1-\theta_y)$ has only been enlarged to one. Likewise the corridor spectrum gives the exact row formula

$$N_R^{\leq}(E) = \sum_{m=1}^{\lfloor r \rfloor} \left[4f_r(m) + 1 - \{4f_r(m)\}\right],$$

and hence

$$N_R^{\leq}(E) \leq \pi r^2 - r + 4c\sqrt{r} - F_R. \tag{4.11}$$

The two sums F_H, F_R have exactly the phases covered by (4.7).

Subtracting the local area budget from the sum of (4.10) and $d/2$ times (4.11) bounds the excess by

$$\left(\frac{d}{2}-1\right)r + c(\sqrt{2}+2d)\sqrt{r} + 1 - F_H - \frac{d}{2}F_R.$$

After (4.7), this is at most

$$\left(\frac{831d}{2000} - \frac{38}{25}\right)r + c(\sqrt{2}+2d)\sqrt{r} + 13 + 5d.$$

Every coefficient increases with d. Thus for $0 \le d \le 3$, using $c < 3/11$ and $\sqrt{2} < 10/7$, the excess is bounded above by

$$-\frac{547}{2000}r + \frac{156}{77}\sqrt{r} + 28.$$

At $r = 800$, the bound $\sqrt{800} < 29$ makes this at most $-50838/385 < 0$. After division by r, both positive terms decrease. We have therefore proved the common tail

$$\begin{aligned} N^{\le}_{\widetilde{C}_S}(\pi^2r^2) + \frac{d}{2}N^{\le}_{\widetilde{R}}(\pi^2r^2) &\le N^{\le}_{\widetilde{H}_S}(\pi^2r^2) + \frac{d}{2}N^{\le}_{\widetilde{R}}(\pi^2r^2) \\ &\le \left(1+\frac{d}{2}\right)\pi r^2 \qquad (r \ge 800,\ d \le 3). \end{aligned} \tag{4.12}$$

Only form-domain inclusion was used in replacing C_S by H_S.

The four-port chamber requires larger blocks. The same exact lower-sum construction, with the parameters specified in Appendix A, gives

$$\begin{aligned} A(1,13,3) &\ge \frac{37632015538794073}{106496000000000000} > \frac{353}{1000}, \\ A(4,7,8) &\ge \frac{32022499598134509}{114688000000000000} > \frac{279}{1000}. \end{aligned} \tag{4.13}$$

Thus (4.3) gives, for the same chamber and corridor fractional sums,

$$F_H \ge \frac{353}{500}r - \frac{559}{8}, \qquad F_R \ge \frac{279}{1000}r - \frac{105}{2}.$$

The degree-four specialization of the preceding row bookkeeping bounds the excess $N^{\le}_{\widetilde{H}}(\pi^2r^2) + 2N^{\le}_{\widetilde{R}}(\pi^2r^2) - 3\pi r^2$ by

$$r + c(\sqrt{2}+8)\sqrt{r} + 1 - F_H - 2F_R \le -\frac{33}{125}r + \frac{18}{7}\sqrt{r} + \frac{1407}{8},$$

where $c(\sqrt{2}+8) < 18/7$. At $r = 1024$ this last expression equals $-85227/7000 < 0$. Again, after division by r, the positive terms decrease. If C_4 denotes the actual four-port chamber and H its fully Neumann relaxation, we obtain

$$\begin{aligned} N^{\le}_{\widetilde{C}_4}(\pi^2r^2) + 2N^{\le}_{\widetilde{R}}(\pi^2r^2) &\le N^{\le}_{\widetilde{H}}(\pi^2r^2) + 2N^{\le}_{\widetilde{R}}(\pi^2r^2) \\ &\le 3\pi r^2 \qquad (r \ge 1024). \end{aligned} \tag{4.14}$$

The finite interval for this last chamber will be completed after the five degree-at-most-three types.

5. The finite certificate for degrees at most three

The remaining interval is finite, but its verification must still retain all multiplicities and all real energy thresholds. In the coordinate $P = 16r^2$, every separated chamber event is $P = a^2+b^2$, with the five numerator pairs listed in Section 2. Corridor events are $P = a^2+16b^2$, with $a \ge 0$ and $b \ge 1$ integers. Repeated representations are distinct eigenfunctions and are counted separately.

Write $N_H = N^{\le}_{\widetilde{H}_S}(\pi^2P/16)$ and $N_R = N^{\le}_{\widetilde{R}}(\pi^2P/16)$ when the energy argument is suppressed. The exact rational comparison

$$16 \cdot 33102\,(2N_H + dN_R) \le 103993(2+d)P \tag{5.1}$$

implies the desired local estimate because $\pi > 103993/33102$. Scanning every union event through $P_* = 16 \cdot 800^2 = 10240000$, its complete list of failures before any boundary correction is

type	P	N_H	N_R
$d = 2$, adjacent	2	1	0
$d = 2$, opposite	20	5	3
$d = 3$	1	1	0
$d = 3$	25	7	4

The scan is specified completely in Appendix A. This table locates the needed boundary information; the corrected verification below, including the correction endpoints, is what proves the finite interval.

Use the actual-cell boundary estimates (3.1), (3.6), and (3.23): the ground level is greater than $\pi^2/6$; the indicated opposite-port sector level is greater than $41\pi^2/32$; and the two degree-three cluster levels are greater than $53\pi^2/32$. Define the number of removed chamber states by

$$\begin{aligned} &c_S(P) = 0, && d = 0, 1, \\ &c_S(P) = \mathbf{1}_{[20,41/2)}(P), && d = 2, \text{ opposite}, \\ &c_S(P) = \mathbf{1}_{\{P<8/3\}} \min\{1, N_H\}, && d = 2, \text{ adjacent}, \\ &c_S(P) = \mathbf{1}_{\{P<8/3\}} \min\{1, N_H\} + 2\mathbf{1}_{[25,53/2)}(P), && d = 3. \end{aligned} \tag{5.2}$$

The ground correction is used only after its relaxed state has appeared; there is exactly one such state in the stated interval. The sector comparisons, with the other sectors left at their relaxed bounds, give

$$N_{\overline{C}_S}^{\leq}(\pi^2 P/16) \leq N_H - c_S(P).$$

All left endpoints in (5.2) are included. At the three right endpoints the corrections are conservatively set to zero, even though the spectral lifts are strict there as well.

For the corrected scan, replace $2N_H + dN_R$ in (5.1) by

$$U_S(P) = 2\big(N_H - c_S(P)\big) + dN_R,$$

and record the integer or rational margin

$$\mathcal{M}_S(P) = 103993(2 + d)P - 16 \cdot 33102\, U_S(P). \tag{5.3}$$

Every actual union event through P_* has positive margin. The complete event totals and minimum margins are

type	actual union events	minimum margin	at P
$d = 0$	1028343	604624	8
$d = 1$	1390540	500631	5
$d = 2$, opposite	1028657	440576	80
$d = 2$, adjacent	1402109	222632	26
$d = 3$	1391119	198901	89

These totals include corridor events when $d = 0$; including them is harmless and makes the enumeration rule uniform. The additional correction-reset events give

type	P	$U_S(P)$	$\mathcal{M}_S(P)$
$d = 2$, opposite	41/2	16	53314
$d = 2$, adjacent	8/3	2	149984/3
$d = 3$	8/3	2	981928/3
$d = 3$	53/2	26	17281/2

Since all actual events have integer P, the counts at these noninteger thresholds equal the cumulative counts at $\lfloor P \rfloor$. The corrections are evaluated at the threshold itself, so they have already switched off.

Between two consecutive actual or reset events the corrected upper count is constant, while the area budget increases with P. Before the first event it is zero. This proves the local

comparison for every real $0 < P \le P_*$, not just at separated eigenvalues. The endpoint P_* itself is a corridor event and is included in the scan; it also belongs to (4.12). Thus the finite and infinite ranges join without a missing energy, and together yield

$$N^{\le}_{C_S}(E) + \frac{d}{2} N^{\le}_{R}(E) \le \frac{(4+2d)E}{4\pi} \qquad (E > 0,\ d \le 3). \tag{5.4}$$

This completes five of the six local comparisons. The next section supplies the four-port estimate and then sums the local bounds over the physical network, including all interface shares and multiplicities.

6. The four-port chamber

The remaining chamber has four centered Neumann windows and retains Dirichlet boundary at all four corners. Denote it by C_4, and denote the fully Neumann square by H. The estimate to be proved is

$$N^{\le}_{C_4}(E) + 2N^{\le}_{R}(E) \le \frac{3E}{\pi} \qquad (E > 0). \tag{6.1}$$

The fully Neumann comparison alone has several low-energy deficits. We repair these by retaining moments of the actual Dirichlet trace. The resulting matrices have order at most six; their complements are bounded analytically, rather than omitted.

6.1. **A finite block with a complete spectral complement.** Put

$$X_m(x) = \cos\frac{m\pi(x+1)}{2}, \qquad \Psi_{mn}(x,y) = X_m(x)X_n(y), \qquad q_{mn} = \frac{m^2+n^2}{4},$$

for $m, n \ge 0$, and let $d_0 = 2$, $d_m = 1$ for $m \ge 1$. Thus the unnormalized Neumann basis has squared norms $\nu_{mn} = d_m d_n$ and energies $\pi^2 q_{mn}$. Let Γ be the union of the Dirichlet portions of the four faces. On each face its tangential coordinate belongs to $[-1,-1/2] \cup [1/2,1]$.

Fix $e > 0$ outside the Neumann spectrum. From the modes with $q_{mn} < e$ select k mutually orthogonal vectors $g_1, \dots, g_k$, each contained in a single Neumann eigenspace, of respective dimensionless energies $q_1, \dots, q_k$. Write M for the number of all Neumann modes below e, including multiplicity. Impose orthogonality to the remaining $M - k$ low directions. Every function in this constrained actual form domain then has the expansion

$$u = \sum_{j=1}^{k} a_j g_j + h, \qquad h \in \overline{\operatorname{span}}\{\Psi_{mn} : q_{mn} > e\}.$$

Define the boundary moments and matrices by

$$L_i(v) = \int_\Gamma \overline{g_i}\ \operatorname{Tr} v\ \mathrm{d}s, \qquad D_{ij} = L_i(g_j),$$
$$W_{ij} = \delta_{ij}(e - q_j)\|g_j\|^2_{L^2(C)}, \qquad B_{mn,i} = L_i(\Psi_{mn}),$$
$$S_e = \sum_{q_{mn} > e} \frac{B_{mn}B^*_{mn}}{\nu_{mn}(q_{mn} - e)}.$$

Here D is a boundary Gram matrix and W is positive diagonal. All vectors used below are real; the adjoints permit the same argument for complex form functions. Set

$$Q_e(v) = \pi^{-2}\int_C |\nabla v|^2\ \mathrm{d}x - e\|v\|^2_{L^2(C)}.$$

The trace condition and orthogonality give

$$L(h) = -Da, \qquad Q_e(u) = Q_e(h) - a^* W a.$$

Weighted Cauchy–Schwarz in the complete high-mode expansion yields

$$|z^* Da|^2 \le (z^* S_e z) Q_e(h) \qquad (z \in \mathbb{C}^k). \tag{6.2}$$

The tail bound below proves convergence of S_e. The expansion also converges in H^1, so continuity of the trace justifies this calculation first for partial sums and then for every form function.

Suppose an explicit positive matrix $\overline{S}$ satisfies $S_e \le \overline{S}$ and

$$T = DW^{-1}D - \overline{S} > 0. \tag{6.3}$$

This forces D to be nonsingular: a vector in its kernel would give $z^*Tz = -z^*\overline{S}z \le 0$. Hence $D > 0$ and, by congruence and inversion,

$$D^{-1}\overline{S}D^{-1} < W^{-1}, \qquad K := D\overline{S}^{-1}D > W.$$

Maximizing (6.2) over z gives $Q_e(h) \ge a^*Ka$. There is also a positive Neumann spectral gap $\delta_e = \min_{q_{mn}>e}(q_{mn} - e)$, giving $Q_e(h) \ge \delta_e\|h\|^2$. Choose $\theta > 0$ small enough that $(1-\theta)K - W > 0$. Then

$$Q_e(u) \ge a^*\big((1-\theta)K - W\big)a + \theta\delta_e\|h\|^2.$$

This is uniformly positive relative to $\|u\|^2$ on the constrained subspace. The max–min principle, with precisely $M-k$ orthogonality conditions, therefore proves

$$\lambda_{M-k+1}(C_4) > \pi^2 e, \qquad N^{\le}_{C_4}(\pi^2 e) \le M - k. \tag{6.4}$$

In particular, equality at the target energy is excluded. This is a lower-bound argument on the actual form domain, not an upper-bound Ritz approximation.

6.2. **Explicit entries and the infinite high-mode tail.** The tangential integrals needed for every entry are

$$J_{ab} = \int_{[-1,-1/2]\cup[1/2,1]} X_a(s)X_b(s)\,\mathrm{d}s.$$

Direct integration gives

$$J_{ab} = \begin{cases} 0, & a+b \text{ odd}, \\ 2\left[\dfrac{\sin((a-b)\pi/4)}{(a-b)\pi} + \dfrac{\sin((a+b)\pi/4)}{(a+b)\pi}\right], & a+b \text{ even}. \end{cases} \tag{6.5}$$

The quotient with zero denominator is assigned its limit $1/4$. For equal parity the sine values are $0, 1, -1$, so J_{ab} is a rational number plus a rational multiple of $1/\pi$. For a single selected mode $g_i = \Psi_{ab}$ the four faces give

$$B_{mn,i} = (1 + (-1)^{a+m})J_{bn} + (1 + (-1)^{b+n})J_{am}. \tag{6.6}$$

Linearity gives the entries for combinations of modes, and $D_{ij} = L_i(g_j)$ uses the same formula.

For completeness, let f be any linear combination of the boundary densities $g_i|_\Gamma$, and extend its restriction to each face by zero on the Neumann window. Write A_n for its tangential cosine moment on one face. One-dimensional Parseval gives

$$\sum_{n\ge 0} \frac{|A_n|^2}{d_n} = \int_{-1}^{1} |f|^2\,\mathrm{d}s.$$

In the normalized two-dimensional basis, a single face contributes $A_n/\sqrt{d_m d_n}$ up to a sign. For a cutoff B with $\beta = (B+1)^2 - 4e > 0$, split the omitted pairs into the normal-index tail $m > B$ and the tangential-index tail $n > B$; their overlap may be counted twice. In the first part,

$$\sum_{m>B} \frac{4}{m^2 + n^2 - 4e} \le \frac{4}{1 - 4e/(B+1)^2} \sum_{m>B} \frac{1}{m^2} \le \frac{4}{B(1 - 4e/(B+1)^2)}.$$

In the second part, keeping the half weight of the normal zero mode,

$$4\left(\frac{1}{2\beta} + \sum_{m\ge 1} \frac{1}{m^2 + \beta}\right) \le \frac{2}{\beta} + 4\int_0^\infty \frac{\mathrm{d}t}{t^2 + \beta} = \frac{2}{\beta} + \frac{2\pi}{\sqrt{\beta}}.$$

Apply Parseval in each part, sum over faces, and use $|z_1+z_2+z_3+z_4|^2 \le 4\sum_i |z_i|^2$ for their combined contribution to a mode. Thus the matrix tail satisfies

$$S_e - S_e^{(B)} \le \varepsilon(B,e)D, \quad \varepsilon(B,e) = 4\left[\frac{4}{B(1-4e/(B+1)^2)} + \frac{2}{\beta} + \frac{2\pi}{\sqrt{\beta}}\right], \tag{6.7}$$

where $S_e^{(B)}$ contains every high pair with $0 \le m,n \le B$. This inequality holds for every linear combination of the moments, and therefore in matrix order, not just entrywise.

All matrix targets below have $e \le 3413/32$. With $B = 128$ we have $\beta > 127^2$, and $\pi < 22/7$ gives

$$\varepsilon(128,e) < \frac{7536117}{23063327} < \frac{1}{3}. \tag{6.8}$$

Consequently it is enough to verify

$$T = DW^{-1}D - S_e^{(128)} - \frac{1}{3}D > 0. \tag{6.9}$$

Indeed, this inequality first forces D to be nonsingular, and then $\overline{S} = S_e^{(128)} + D/3 > 0$, as required above. No part of the positive infinite complement has been dropped.

6.3. **Twenty-eight block certificates.** Continue to use $P = 16E/\pi^2$. For an endpoint P_* in the following table, set $e = P_*/16$ and select all modes satisfying $s \le m^2+n^2 < 4e$. Apart from the specified symmetric row, these individual modes are the block vectors, ordered by m and then n. The columns k and c give the block dimension and the resulting cap $N_{C_4}^{\le}(\pi^2 P_*/16) \le c$.

P_*	s	k	c	P_*	s	k	c
2	0	1	0	371/2	45	2	41
11/2	1	2	1	214	53	2	50
41/2	4	3	5	551/2	68	2	62
55/2	5	2	6	294	73	2	65
65/2	8	1	8	595/2	74	2	67
85/2	10	2	11	683/2	85	4	75
70	17	2	17	428	106	2	94
75	18	1	19	438	109	2	96
85	20	2	20	469	117	2	102
102	25	4	22	603	149	2	131
211/2	26	2	26	834	208	2	179
323/2	40	2	37	2221/2	277	2	234
337/2	40	4	37	1365	340	4	286
343/2	41	2	39	3413/2	425	6	353

For $P_* = 41/2$ use instead the three orthogonal vectors

$$g_1 = \Psi_{02} + \Psi_{20}, \qquad g_2 = \Psi_{12}, \qquad g_3 = \Psi_{21}.$$

Their squared norms are $4,1,1$ and their energies are $1, 5/4, 5/4$. The discarded antisymmetric vector $\Psi_{02} - \Psi_{20}$ is one of the five low orthogonality constraints. In particular this row removes three directions, not four.

Every endpoint avoids the Neumann spectrum. Substitution in (6.9), using outward-rounded rational intervals with denominator 2^{48}, gives positive symmetric-elimination pivots in all 28 cases. Their lower endpoints all exceed $3/4$; the smallest is

$$\frac{111857158525441}{140737488355328} > \frac{3}{4}, \qquad P_* = 1365. \tag{6.10}$$

Appendix B specifies every arithmetic operation and loop needed to check this assertion. Thus the table follows from (6.4), with strict spectral lifts at its endpoints.

6.4. **One additional odd–odd direction.** The square and its boundary conditions are invariant under reflection in both coordinate axes. The corresponding four parity subspaces are orthogonal for both mass and energy. Here an odd index m means that X_m is odd under $x \mapsto -x$. In the odd–odd subspace the first four relaxed modes are

$$\Psi_{11}, \quad \Psi_{13}, \quad \Psi_{31}, \quad \Psi_{33},$$

with dimensionless energies $1/2, 5/2, 5/2, 9/2$; the next energy is $13/2$. We shall prove

$$\lambda_4(C_4; \text{odd–odd}) > \frac{163\pi^2}{32}. \tag{6.11}$$

On a face use $t = (s+1)/2$ and the density

$$f(t) = -\frac{7}{20}\cos(\pi t) + \cos(3\pi t) - \frac{53}{100}\cos(5\pi t)$$

on $[0, 1/4] \cup [3/4, 1]$, with value zero on the remaining interval. Use this density with positive signs on the left and bottom faces and negative signs on the right and top faces. Let L be the resulting boundary integral, and put

$$(a_1, a_3, a_5) = \left(-\frac{7}{20}, 1, -\frac{53}{100}\right), \qquad F_n = \sum_{j\in\{1,3,5\}} a_j J_{jn},$$
$$b = 4F_3, \qquad D_f = 4 \sum_{i,j\in\{1,3,5\}} a_i a_j J_{ij}.$$

The measure in J_{ij} is physical arclength $\mathrm{d}s = 2\,\mathrm{d}t$. For odd m, n, all Ψ_{mn} have norm one and $L(\Psi_{mn}) = 2(F_m + F_n)$; on the other parity sectors the moment vanishes. At $e = 163/32$ its complete high-mode Gram sum is

$$S = \sum_{\substack{m,n\geq 1 \text{ odd} \\ q_{mn} > e}} \frac{4(F_m + F_n)^2}{q_{mn} - e}.$$

The finite sum with $m, n \leq 256$ contains exactly 16380 high modes. The general tail estimate (6.7) applies to this density as well. Since $\beta = 257^2 - 4e = 528229/8 > 256^2$, it gives

$$S - S^{(256)} \leq \frac{9516431}{59161648} D_f.$$

Exact outward rounding, described in Appendix B, gives

$$b \geq \frac{5689200238183}{12500000000000} > 0, \qquad D_f \leq \frac{21153037117543}{100000000000000},$$
$$S^{(256)} \leq \frac{29181627575971}{100000000000000}, \qquad S - S^{(256)} \leq \frac{3402566104473}{100000000000000}.$$

Therefore, with $\overline{S} = 8146048420111/25000000000000$,

$$\frac{32}{19}b^2 - \overline{S} \geq \frac{213752279113583986864148 9}{92773437500000000000000000} > \frac{1}{50}. \tag{6.12}$$

To see explicitly what this certifies, impose odd–odd orthogonality to $\Psi_{11}, \Psi_{13}, \Psi_{31}$ and write $u = a\Psi_{33} + h$. The high part has $Q_e(h) \geq (45/32)\|h\|^2$, while $L(h) = -ab$ and $Q_e(u) = Q_e(h) - (19/32)|a|^2$. Weighted Cauchy–Schwarz gives $|a|^2 b^2 \leq \overline{S} Q_e(h)$. With $\rho = (19/32)\overline{S}/b^2 < 1$,

$$Q_e(u) \geq (1-\rho) Q_e(h), \qquad \|u\|^2 \leq \left(\frac{\overline{S}}{b^2} + \frac{32}{45}\right) Q_e(h).$$

This is a uniform positive form gap under three constraints and proves (6.11).

The table row $P_* = 85$ uses precisely Ψ_{24} and Ψ_{42}. Their moments couple only to the even–even sector, so the same matrix proof removes two states within that sector through $P = 85$. At $P = 163/2$ the fully Neumann square has 22 states, of which exactly four are

odd–odd. The odd–odd bound removes one of them, the even–even bound removes two others, and all remaining sectors keep their Neumann upper counts. Thus

$$N^{\leq}_{\widetilde{C}_4}\left(\frac{163\pi^2}{32}\right) \leq 19. \tag{6.13}$$

Monotonicity makes this a cap at every smaller energy as well.

6.5. **All events, all resets, and the global conclusion.** Let $\mathcal{C}$ consist of the 28 endpoint–cap pairs in the table and the pair $(163/2, 19)$. Write

$$H(P) = N^{\leq}_{\widetilde{H}}(\pi^2 P/16), \qquad R(P) = N^{\leq}_{\widetilde{R}}(\pi^2 P/16),$$

and use the upper step function

$$U(P) = \min\big(\{H(P)\} \cup \{c : (p, c) \in \mathcal{C},\ P < p\}\big). \tag{6.14}$$

Each cap is conservatively stopped at its own endpoint, even though it was proved inclusively. The actual chamber count is at most $U(P)$. Chamber events are $P = 4(m^2+n^2)$ with $m, n \geq 0$; corridor events are $P = m^2 + 16n^2$ with $m \geq 0$, $n \geq 1$. Include their multiplicities before each comparison, and add all cap endpoints and 1707 to the event list. There are exactly 300 distinct events in $[0, 1707]$. At every such event the exact rational test gives

$$3 \cdot 103993P - 16 \cdot 33102\big(U(P) + 2R(P)\big) \geq 0. \tag{6.15}$$

The margin at zero is zero. The smallest positive-energy margin is 291, at $P = 73$, where $U(P) = 19$ and $R(P) = 12$. Appendix B gives the complete finite enumeration. Between consecutive events both upper counts are constant and the budget increases. Since $\pi > 103993/33102$, this proves (6.1) throughout $0 < P \leq 1707$, including every reset and every eigenvalue equality.

The bridge from this interval to the analytic tail is also exact. Enumerating all separated union events through 2^{24} gives 1657038 events and only the 30 deficits listed in Appendix B; their largest coordinate is 1700. If a deficient event has value $B = H(P) + 2R(P)$, its constant count is paid by the rational budget once

$$P \geq \frac{176544B}{103993}.$$

All 30 such coordinates are below 1707, the largest being $177426720/103993 < 1707$. Consequently the uncorrected comparison already holds at every real $1707 \leq P \leq 2^{24}$. At $P = 2^{24}$ we have $r = 1024$, so (4.14) covers the whole remaining range with the same endpoint. This proves (6.1) at every energy.

Together with (5.4), the local budget (2.10) now holds for all six chamber types, without a degree restriction beyond the four available faces. For the network in Section 2, Neumann cutting after dilation by two and the identity $\sum_{v\in V} d(v) = 2|F|$ give

$$N^{\leq}_{2\Omega}(E) \leq \sum_{v\in V} N^{\leq}_{\widetilde{C}_{S_v}}(E) + |F| N^{\leq}_{\widetilde{R}}(E) \leq \frac{4(|V| + |F|)E}{4\pi}.$$

Returning to the original scale proves $N^{\leq}_{\Omega}(\mathcal{E}) \leq |\Omega|\mathcal{E}/(4\pi)$ for every $\mathcal{E} > 0$, and hence $\lambda_k(\Omega) \geq 4\pi k/|\Omega|$ for every $k \geq 1$. The argument permits every finite choice of vertices and selected nearest-neighbour edges in the specified geometry; its constants establish the compact comparison used in Section 7 to obtain further width and length ranges.

7. Nonuniform corridors and displaced ports

The compact local inequalities also give a family with variable corridor dimensions. The rooms in this section retain side length two. A corridor $R(L, w)$ has length L, width w,

Neumann conditions at its short ends, and Dirichlet conditions along its long sides. Each physical corridor will satisfy one of the two conditions

$$\begin{aligned} &w = 1, \quad L \in \{4\} \cup [7, \infty), \qquad \text{or} \\ &0 < w \le 1, \quad L \ge 9, \quad Lw \ge 4. \end{aligned} \tag{7.1}$$

Its port on a room face may have center displacement η from the face center, provided

$$|\eta| \le \frac{1-w}{2}. \tag{7.2}$$

Thus the actual port lies inside the centered unit comparison window. The two displacements on a corridor's endpoint rooms may differ whenever the physical geometry realizes them. No reflection symmetry of the actual chamber is required.

Consider any finite assembly of these axis-parallel square rooms and straight rectangular corridors, with disjoint piece interiors and only the designated room–corridor contacts. There is at most one port on each room face. All remaining piece boundary is physical Dirichlet boundary. Writing n for the number of rooms, its area and eigenvalue count satisfy

$$|\Omega| = 4n + \sum_e L_e w_e, \qquad N_\Omega^{\le}(E) \le \frac{|\Omega| E}{4\pi} \quad (E > 0). \tag{7.3}$$

We prove this using only the established compact cell estimates and elementary strip bounds. An explicit grid construction below supplies admissible domains with independent corridor widths and displacements.

7.1. **A strip with sufficient length pays for its own area.** For $L \ge 3$, the unit-width mixed strip satisfies

$$N_{R(L,1)}^{\le}(E) \le \frac{LE}{4\pi}. \tag{7.4}$$

To verify this at every energy, write $E = \pi^2 r^2$, $M = \lfloor r \rfloor$, and $f_r(t) = \sqrt{r^2 - t^2}$. Exact separation gives

$$N_{R(L,1)}^{\le}(\pi^2 r^2) = \sum_{m=1}^{M} \left(\lfloor L f_r(m) \rfloor + 1 \right).$$

The term with $m = r$ is present when r is an integer, and contributes the longitudinal zero mode. By (4.8),

$$N_{R(L,1)}^{\le}(\pi^2 r^2) \le \frac{L\pi r^2}{4} - L\left[\left(\frac{1}{2} - \frac{1}{L} \right) r - c\sqrt{r} \right], \qquad c = \sqrt{2/27}.$$

When $L \ge 3$ and $r \ge 8/3$, the bracket is nonnegative because $1/2 - 1/L \ge 1/6$ and $c\sqrt{r} \le r/6$.

Below this range there are at most two transverse rows. There are none for $r < 1$. For $1 \le r < 2$, completing a square, with $t = r^2$, gives

$$\frac{\pi t}{4} - \sqrt{t-1} \ge \frac{\pi}{4} - \frac{1}{\pi} > \frac{369}{800} > \frac{1}{3}.$$

This pays the one row contribution $1/L \le 1/3$. For $2 \le r < 8/3$, Cauchy–Schwarz and another completed square give

$$\begin{aligned} \sqrt{t-1} + \sqrt{t-4} &\le 2\sqrt{t - \tfrac{5}{2}}, \\ \frac{\pi t}{4} - \sqrt{t-1} - \sqrt{t-4} &\ge \frac{5\pi}{8} - \frac{4}{\pi} > \frac{1077}{1600} > \frac{2}{3}. \end{aligned}$$

The last quantity pays both row contributions. The rational comparisons use $\pi > 25/8$. These three ranges prove (7.4), including all transition energies. By dilation, a strip $R(L, w)$ pays for its own area whenever $L/w \ge 3$.

If $L = 4 + s$ with $s \ge 3$, one transverse Neumann cut gives

$$N_{R(L,1)}^{\le}(E) \le N_{R(4,1)}^{\le}(E) + N_{R(s,1)}^{\le}(E) \le N_{R(4,1)}^{\le}(E) + \frac{sE}{4\pi}.$$

Thus a unit-width corridor of length at least seven can replace the base corridor while paying for its additional area. No cut is needed when its length is exactly four. This proves the corridor comparison needed for the first range in (7.1).

7.2. **Comparing the area left by different corridors.** Put $E = \pi^2 r^2$ and define the difference between a corridor's area budget and its count by

$$\delta_{L,w}(r) = \frac{\pi L w r^2}{4} - N^{\leq}_{R(L,w)}(\pi^2 r^2), \qquad \delta_0(r) = \delta_{4,1}(r).$$

For a chamber with $d \geq 1$ actual ports, let C_S be the original centered-window comparison chamber. By (7.2), the actual chamber form domain is contained in that of C_S. The compact local inequalities (5.4) and (6.1) therefore bound its excess above the room area by $d\delta_0(r)/2$. The fully relaxed chamber bound (4.10), with its nonnegative fractional loss discarded, also bounds this excess by

$$B_d(r) = (d-1)r + c\sqrt{2r} + 1.$$

Consequently it is enough that each incident corridor satisfy

$$\delta_{L,w}(r) \geq \min\left\{\delta_0(r), \frac{2B_d(r)}{d}\right\}. \tag{7.5}$$

Indeed, the actual chamber excess is at most $d/2$ times the minimum on the right. Adding one half of the d corridor differences then pays that excess. Different incident corridors may satisfy the two alternatives for different reasons. No common width or common activity threshold is required, and the argument does not require either comparison difference to have a prescribed sign.

The first range in (7.1) has $\delta_{L,1} \geq \delta_0$ by the preceding cut argument. Now take $0 < w \leq 1$, $L \geq 9$, $Lw \geq 4$, and set $T = wr$. If $T < 1$, this corridor has no transverse mode, and

$$\delta_{L,w}(r) = \frac{\pi L w r^2}{4} \geq \pi r^2 \geq \delta_0(r).$$

This verifies (7.5) for every inactive corridor.

For $T \geq 1$, define the quarter-circle sum deficit

$$D(T) = \frac{\pi T^2}{4} - \sum_{m=1}^{\lfloor T \rfloor} \sqrt{T^2 - m^2}.$$

The exact row formula at width w gives

$$\delta_{L,w}(r) \geq \frac{L}{w} D(T) - \lfloor T \rfloor, \qquad w\delta_{L,w}(r) \geq L D(T) - \lfloor T \rfloor.$$

Moreover

$$wB_d(r) = (d-1)T + c\sqrt{2wT} + w \leq (d-1)T + c\sqrt{2T} + 1.$$

It therefore suffices to prove, simultaneously for $1 \leq d \leq 4$,

$$9D(T) \geq \lfloor T \rfloor + F_d(T), \qquad F_d(T) = \frac{2}{d}\big((d-1)T + c\sqrt{2T} + 1\big). \tag{7.6}$$

7.3. **The scalar comparison for active corridors.** For fixed d, $F_d(T)$ increases with T. For fixed T, its maximum over $1 \leq d \leq 4$ occurs at $d = 1$ or $d = 4$, since

$$F_d(T) = 2T + \frac{2}{d}(c\sqrt{2T} + 1 - T).$$

These observations reduce (7.6) to four short ranges.

If $1 \leq T \leq 3/2$, the one-row estimate above gives $D(T) > 369/800$. At the upper endpoint, the two possible bounds for $F_d(T) + 1$ are

$$3 + \frac{2\sqrt{2}}{3} < 4, \qquad \frac{15}{4} + \frac{\sqrt{2}}{6} < 4.$$

Both use $\sqrt{2} < 3/2$, whereas $9D(T) > 3321/800 > 4$.

If $3/2 \le T < 2$, the function $\pi T^2/4 - \sqrt{T^2-1}$ is increasing: its derivative is $T(\pi/2 - 1/\sqrt{T^2-1}) > 0$ there. Hence

$$D(T) \ge \frac{9\pi}{16} - \frac{\sqrt{5}}{2} > \frac{225}{128} - \frac{9}{8} = \frac{81}{128}.$$

Here $\pi > 25/8$ and $\sqrt{5} < 9/4$. At $T = 2$, the two bounds for $F_d + 1$ are $3 + 4c$ and $9/2 + c$, both less than five because $c < 1/2$. Since $9D(T) > 729/128 > 5$, the comparison follows on this range too.

If $2 \le T < 8/3$, write $t = T^2$ and use weighted Cauchy–Schwarz:

$$\sqrt{t-1} + \sqrt{t-4} \le \sqrt{3\left(\frac{t-1}{2} + t - 4\right)} = \frac{3}{\sqrt{2}}\sqrt{t-3}.$$

Completing the square now yields the stronger two-row estimate

$$D(T) \ge \frac{3\pi}{4} - \frac{9}{2\pi} > \frac{723}{800}.$$

For $T \ge 2$, $c\sqrt{2T} + 1 \le T$, so the maximum over d occurs at $d = 4$. At $T = 8/3$ this gives

$$F_d(T) + 2 \le \frac{13}{2} + \frac{2\sqrt{2}}{9} < 7, \qquad 9D(T) > \frac{6507}{800} > 8.$$

The inclusion of the second zero-radius row at $T = 2$ is reflected by the term $+2$ throughout this range.

Finally, if $T \ge 8/3$, (4.8) gives $D(T) \ge T/2 - c\sqrt{T} \ge T/3$. The elementary inequality $c\sqrt{2T} + 1 \le T$ follows, for example, by separately bounding both terms by $T/2$ when $T \ge 2$. Therefore

$$9D(T) - \lfloor T \rfloor \ge 2T \ge F_d(T).$$

This proves (7.6) at every active threshold, and hence (7.5) for every corridor in (7.1).

7.4. **Exact area and an explicit realization.** For each chamber, assign half the area and half the count of each incident corridor. The preceding minimum comparison gives

$$N^{\le}_{C_v^{\mathrm{actual}}}(E) + \frac{1}{2}\sum_{e\sim v} N^{\le}_{R(L_e,w_e)}(E) \le \frac{E}{4\pi}\left(4 + \frac{1}{2}\sum_{e\sim v} L_e w_e\right).$$

An isolated room is covered by the degree-zero local estimate. Summing over all rooms counts each selected edge twice with weight one half. Neumann cutting at the actual interfaces then proves (7.3). No area is assigned to a missing edge, and the argument is unchanged by cycles or disconnected components.

Here is a concrete construction. Choose increasing finite coordinate lists (X_i) and (Y_j) whose successive gaps belong to $\{6\} \cup [9, \infty)$. Select any finite set of grid vertices and any nearest-index-neighbour edges with selected endpoints. Place a side-two room at each selected point (X_i, Y_j). A horizontal edge across a gap $g = X_{i+1} - X_i$ has length $L = g - 2$ and corridor

$$[X_i + 1, X_{i+1} - 1] \times [Y_j + \eta_e - w_e/2, Y_j + \eta_e + w_e/2].$$

Use the coordinate-swapped expression for a vertical edge. At gaps $g = 6$ or $9 \le g < 11$, take $w_e = 1$, $\eta_e = 0$. At any gap $g \ge 11$, one may independently choose

$$\frac{4}{g-2} \le w_e \le 1, \qquad |\eta_e| \le \frac{1 - w_e}{2}.$$

Thus all edges across a coordinate band share its geometric length, while their widths and transverse displacements can be chosen independently. The same displacement is used at the two ends of that straight corridor; different edges incident to a room need not have the same displacement.

Every horizontal corridor lies in a gap between room columns and within the centered unit band of one room row. Every vertical corridor lies in a gap between rows and within the corresponding unit band of a column. Since a room extends one unit from its center, while

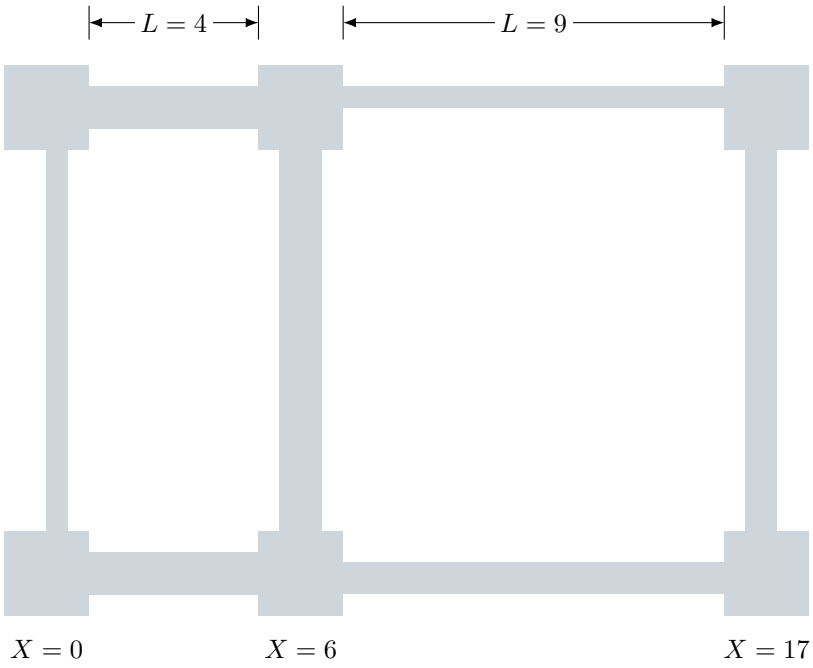


FIGURE 3. An actual nonuniform network, with side-two room centers at $\{0, 6, 17\} \times \{0, 11\}$. The left horizontal passages have $(L, w, \eta) = (4, 1, 0)$. The right passages have $(9, 3/4, -1/8)$ below and $(9, 1/2, 1/4)$ above. The vertical passages, from left to right, have $(9, 1/2, 1/4)$, $(9, 1, 0)$, and $(9, 3/4, -1/8)$. Horizontal offsets are measured upwards and vertical offsets to the right. All lengths and widths are drawn at the same scale.

these bands extend only one half-unit, perpendicular corridors cannot cross in their interiors. Distinct parallel corridors and nonincident rooms are separated by the coordinate gaps. The only open contacts are the listed centered-envelope ports. This proves admissibility directly for every selected graph and every stated parameter choice.

For example, on a grid of spacing eleven, every selected corridor has length nine. Its width may be chosen independently in $[4/9, 1]$. A width-$1/2$ corridor may be displaced by any amount up to $1/4$ from the face center. After scaling rooms back to side one, this gives width $1/4$, length $9/2$, and displacement up to $1/8$. The resulting families allow degree-four branching and arbitrary finite selected-edge patterns as well as these continuous geometric parameters.

The allowed range also contains passages of fixed area and arbitrarily small width: take $0 < w \leq 4/9$ and $L = 4/w$. Then $L \geq 9$ and $Lw = 4$, the area of a base corridor. Such passages can be realized by choosing the corresponding coordinate gap $2 + 4/w$; different bands may have different gaps. This describes a genuine parameter family, not an assertion that arbitrary prescribed edge lengths satisfy the closure constraints of every cyclic drawing.

The restrictions in (7.1) and (7.2) are part of the result. In particular, the argument gives neither arbitrary width perturbations at length four nor displaced full-width ports. It uses the full physical corridor areas and their counting differences to pay the chamber comparison; it does not assume monotonicity of the corridor count divided by area.

8. Independent port displacements and corridor subdivision

The compact network result keeps each port centered. A second cell inequality permits an independent displacement at each end of a passage. The two applications use the same distinction: symmetry may simplify the comparison problem even when the physical domain has no symmetry.



FIGURE 4. An exact offset-chain geometry with four chambers. The chamber bottoms are $0, 1/5, 1/10, 3/10$ and passage bottoms are $11/20, 3/5, 17/25, 4/5$. All incoming and outgoing offsets satisfy (8.2). The drawing uses equal horizontal and vertical units; neither passage lengths nor vertical displacements are exaggerated.

8.1. **An offset-chain family.** Let $B = (0,1) \times (0,2)$. Give it Neumann conditions on the two segments $\{0,1\} \times (1/3, 5/3)$ and Dirichlet conditions on the rest of the boundary. Continue to write R for the 4-by-1 rectangle with Neumann ends and Dirichlet long sides. We will prove

$$N_B(E) + N_R(E) \le \frac{6E}{4\pi} \qquad (E > 0). \tag{8.1}$$

The enlarged windows of B contain every unit-length interval whose lower endpoint lies in $[1/3, 2/3]$.

For $n \ge 1$, choose real numbers c_j, a_j, $0 \le j < n$, with

$$\frac{1}{3} \le a_j - c_j \le \frac{2}{3} \quad (0 \le j < n), \qquad \frac{1}{3} \le a_j - c_{j+1} \le \frac{2}{3} \quad (0 \le j < n-1). \tag{8.2}$$

Define the open chain as the interior of

$$\bigcup_{j=0}^{n-1} \left([5j, 5j+1] \times [c_j, c_j + 2] \;\cup\; [5j+1, 5j+5] \times [a_j, a_j + 1] \right). \tag{8.3}$$

Its area is $6n$. Cutting each actual interface and enlarging the allowed chamber traces to those of B gives

$$N_\Omega(E) \le n\big(N_B(E) + N_R(E)\big) \le \frac{6nE}{4\pi}.$$

At the two physical ends a Dirichlet boundary is only relaxed, never replaced by a stronger boundary condition. Thus (8.1) proves Pólya's inequality for every chain (8.3), including all endpoint equalities in (8.2).

The incoming and outgoing offsets at a chamber need not be equal. Specifying them successively determines the vertical drift of the chambers; the disjoint horizontal slabs prevent unintended intersections. No bound on the number of chambers is used.

The one-chamber members also admit a direct inheritance argument. After translation such a member is

$$S_\delta = \operatorname{int}\big([0,1] \times [0,2] \;\cup\; [1,5] \times [\delta, \delta+1]\big), \qquad \frac{1}{3} \le \delta \le \frac{2}{3}.$$

Its image under $(x, y) \mapsto (6 - x, y + 1)$ has disjoint interior. Together the two copies form a parent whose vertical sections, away from the joining lines, are $[0, 2]$ for $0 < x < 1$, $[\delta, \delta + 2]$ for $1 < x < 5$, and $[1, 3]$ for $5 < x < 6$. The vertical translates of this parent by $2\mathbb{Z}$ tile a strip, so congruent-subtile inheritance also proves Pólya's inequality for S_δ; see [2, 5]. The cell proof below treats the entire independently displaced chain family.

8.2. **The two boundary corrections.** Relax both vertical faces of B completely to Neumann and call the resulting problem B_0. At $E = \pi^2 r^2$ its spectrum and that of R give the exact sum

$$\begin{aligned} Q_0(r) &= N_{B_0}(\pi^2 r^2) + N_R(\pi^2 r^2) \\ &= \lfloor 2r \rfloor + \lfloor r \rfloor + \sum_{m=1}^{\lfloor r \rfloor} \left(\left\lfloor 2\sqrt{r^2 - m^2} \right\rfloor + \left\lfloor 4\sqrt{r^2 - m^2} \right\rfloor \right). \end{aligned} \tag{8.4}$$

In units of π^2, the individual levels are $m^2/4+n^2$, $m\ge 1, n\ge 0$, for B_0, and $m^2+n^2/16$, $m\ge 1, n\ge 0$, for R. At $r^2=5/4$ the sum is six, exceeding $15\pi/8$; at $r^2=5$ it is twenty-four, exceeding $15\pi/2$.

The problem B is invariant under reflection in both midlines. Write a minus sign for odd parity and a plus sign for even parity. In the $(-,+)$ sector, use distances from the midlines as coordinates on $Q=(0,1/2)\times(0,1)$. The comparison has Dirichlet condition at $x=0$ and $y=1$, and Neumann condition at $x=1/2$ and $y=0$. The actual problem adds Dirichlet condition on $\{1/2\}\times(2/3,1)$. Its normalized ground mode and data for (3.4) are

$$\phi_1=\sqrt{8}\sin(\pi x)\cos(\pi y/2),\qquad \lambda_0=5\pi^2/4,\quad \Delta=2\pi^2,\quad g_1=\frac{4}{3}-\frac{2\sqrt{3}}{\pi}.$$

The trace formula follows by integrating $8\cos^2(\pi y/2)$ over $(2/3,1)$. The resulting bound is

$$\lambda_1^{-,+}(B)\ge \frac{5\pi^2}{4}+\frac{2\pi^2 g_1}{g_1+\pi\sqrt{13}}>\frac{41\pi^2}{32}. \tag{8.5}$$

Indeed $g_1>16/75$, using $\pi>25/8$ and $\sqrt{3}<7/4$; therefore $63g_1>336/25>88/7>\pi\sqrt{13}$, which is the required strict inequality.

In the $(-,-)$ sector the quarter comparison has Dirichlet condition at $x=0$ and $y=0,1$, and Neumann condition at $x=1/2$. Its first three levels are $2\pi^2, 5\pi^2, 10\pi^2$. Apply (3.4) at the second, simple level, imposing orthogonality to the comparison ground mode. The data are

$$\phi_2=\sqrt{8}\sin(\pi x)\sin(2\pi y),\qquad \lambda_0=5\pi^2,\quad \Delta=5\pi^2,\quad g_2=\frac{4}{3}+\frac{\sqrt{3}}{2\pi}.$$

Thus

$$\lambda_2^{-,-}(B)\ge 5\pi^2+\frac{5\pi^2 g_2}{g_2+2\pi\sqrt{10}}>\frac{21\pi^2}{4}. \tag{8.6}$$

For the last comparison, $19g_2>76/3>176/7>2\pi\sqrt{10}$. Multiplicity of the next level causes no problem: the gap above the simple second level is still $5\pi^2$.

All other levels remain bounded below by their comparison levels. Consequently N_B+N_R is bounded by Q_0-1 when

$$40\le 32r^2<41\qquad\text{or}\qquad 160\le 32r^2<168, \tag{8.7}$$

and by Q_0 elsewhere. The upper endpoints are deliberately excluded from the correction, though the strict spectral bounds would allow them. Their upper-count jumps will be tested directly.

8.3. **A short explicit tail for the chain.** Put $f_r(y)=\sqrt{r^2-y^2}$, $M=\lfloor r\rfloor$ and $s=r-M$. Concavity, the trapezoidal inequality, and the chord on the final interval give

$$\begin{aligned}\sum_{m=1}^{M} f_r(m)&\le \frac{\pi r^2}{4}-\frac{r}{2}+\frac{f_r(M)}{2}-\int_M^r f_r(y)\,\mathrm{d}y\\ &\le \frac{\pi r^2}{4}-\frac{r}{2}+\frac{(1-s)\sqrt{2rs}}{2}\le \frac{\pi r^2}{4}-\frac{r}{2}+\frac{\sqrt{2r}}{3\sqrt{3}}.\end{aligned} \tag{8.8}$$

Define

$$F(r)=\sum_{m=1}^{M}\left(\{2f_r(m)\}+\{4f_r(m)\}\right).$$

Bounding the zero-frequency rows in (8.4) by $3r$ and using (8.8) yields

$$Q_0(r)\le \frac{3\pi r^2}{2}+2\sqrt{\frac{2}{3}}\sqrt{r}-F(r). \tag{8.9}$$

A two-row version of the fractional-block argument suffices here. For $a=2,4$, the magnitude of the derivative of af_r at $y=rx$ is $ax/\sqrt{1-x^2}$. On each interval $[p/1000, q/1000]$ in the following table it lies in $[j+1/4, j+3/4]$. These assertions are the integer inequalities

j	(p,q) for $a=2$	(p,q) for $a=4$
0	(125,351)	(63,184)
1	(530,658)	(299,400)
2	(748,808)	(491,566)
3	(852,882)	(631,683)
4	(905,921)	(729,764)
5	(935,944)	(796,820)
6	(953,958)	(843,860)
7	(964,968)	(876,888)
8	(972,974)	(900,909)
9	(978,979)	(918,925)

TABLE 1. Rational derivative windows for the chain count. The intervals are disjoint within each column and their total length over both columns is 934/1000.

$$\begin{aligned}[16a^2+(4j+1)^2]p^2 &\geq (4j+1)^2\,10^6,\\ [16a^2+(4j+3)^2]q^2 &\leq (4j+3)^2\,10^6.\end{aligned} \tag{8.10}$$

An interval of length w in the x variable contains at least $rw/2-1$ disjoint pairs of consecutive integer rows after scaling by r. In each pair the two values of af_r differ by a number whose fractional part lies in $[1/4,3/4]$. Their two fractional parts therefore sum to at least $1/4$: for a difference with fractional part d, the sum is at least $\min(d,1-d)$. Summing the twenty disjoint-window contributions gives

$$F(r) \geq \frac{934}{8000}r - 5. \tag{8.11}$$

For $r \geq 300$, use $2\sqrt{2/3} < 5/3$, $\sqrt{r} \leq r/17$ and $5 \leq r/60$. Since $934/8000 - 1/60 > 5/51$, (8.9) proves

$$Q_0(r) \leq \frac{3\pi r^2}{2} \qquad (r \geq 300). \tag{8.12}$$

8.4. **The remaining finite interval.** Use $X = 32r^2$ for this cell. All comparison events below $r = 300$ are the integers

$$\begin{aligned}X &= 8m^2+32n^2, \quad 1\leq m\leq 600, \quad 0\leq n\leq 300,\\ X &= 32m^2+2n^2, \quad 1\leq m\leq 300, \quad 0\leq n\leq 1200,\end{aligned} \tag{8.13}$$

retaining only $X \leq 2{,}880{,}000$. These bounds are exhaustive by positivity of the summands. Add the artificial events $X = 41, 168$. At each event form the complete inclusive comparison count and subtract one exactly in the intervals (8.7). Call the resulting upper count $U(X)$. The finite integer certificate is

$$6784\,U(X) \leq 999X. \tag{8.14}$$

It suffices because $\pi > 333/106$. Direct enumeration by the algorithm in Appendix A gives 135,424 actual event levels and 135,426 levels after adding the resets. The minimum of $999X - 6784U(X)$ is 255, attained at $X = 41$, where $U(X) = 6$. The comparison count at the cutoff is 423,767. The same verifier checks every inequality in (8.10) and the total window length.

Between successive events U is constant and the area term increases; before the first event the count is zero. Thus the finite certificate proves (8.1) through $r = 300$. Together with (8.12) it proves (8.1) for every energy, completing the offset-chain assertion.

One related boundary convention is useful. Replace the final full passage by two terminal 2-by-1 passages, one at each end of a chain with n chambers and $n-1$ full passages. Each terminal passage has a physical Dirichlet end and a Neumann cut at its chamber. For a transverse row with $s = \sqrt{E/\pi^2 - m^2}$, their combined count is $2\lfloor 2s+1/2\rfloor \leq \lfloor 4s\rfloor + 1$. Indeed,

writing $2s = j + t$ with $0 \le t < 1$ verifies this in the two cases $t < 1/2$ and $t \ge 1/2$. The total terminal count is therefore at most that of one full passage. The area remains $6n$, so the all-energy bound holds for this two-ended variant too.

8.5. **Redistributing an actual corridor budget.** The same local inequality permits a useful geometric dissection. If $w_1, \dots, w_s > 0$ and $\sum_i w_i = 1$, imposing Dirichlet conditions along longitudinal lines in R gives

$$\sum_{i=1}^{s} N_{R(4,w_i)}(E) \le N_R(E). \tag{8.15}$$

Here $R(L, w)$ denotes a rectangle of length L and width w, with Neumann ends and Dirichlet long sides. Extension by zero from the strips gives an isometric embedding of their direct-sum form domain into that of R, proving the direction of (8.15). In contrast, cutting a long passage into k length-four pieces with Neumann cuts gives

$$N_{R(4k,w)}(E) \le k N_{R(4,w)}(E). \tag{8.16}$$

Both comparisons preserve actual area.

Suppose a finite physical assembly has n copies of the chamber underlying B, with every interface on its permitted windows, and rectangular passages that can be cut into length-four pieces. Assume those pieces can be grouped into n groups whose widths each sum to one. All other boundary segments are physical Dirichlet boundary, and piece interiors are disjoint. Cutting the assembly, applying (8.15) in each group, and then (8.1) proves

$$|\Omega| = 6n, \qquad N_\Omega(E) \le \frac{6nE}{4\pi}.$$

This assertion concerns an actual embedded assembly; it does not assert that every abstract graph admits such an embedding. The width grouping is also substantive: a total sum equal to n does not by itself replace the required partition into n unit-width groups. These explicit conditions keep the comparison tied to the physical passage area.

9. What the retained boundary controls

The proofs isolate a local distinction. Fully separated rectangular cells make all frequencies accessible to exact counting, but they can have too many low modes. Retaining the physical Dirichlet segments removes precisely the modes needed in the present geometries. The correction is independent of the size of the assembled graph, which is why arbitrary finite combinatorial complexity causes no accumulation of error.

It is important that some Dirichlet boundary remains. The following direct example explains why independent-cell estimates cannot be uniform over unrestricted port widths. Take a unit-square chamber with four centered Neumann ports of width $w = 1 - 2\varepsilon$, retaining Dirichlet condition on the corner segments. Put $R_0 = 1/4$ and $0 < \varepsilon < R_0$. At each corner choose a function that is zero within distance ε, equals

$$\frac{\log(r/\varepsilon)}{\log(R_0/\varepsilon)} \quad \text{for } \varepsilon < r < R_0,$$

and is one outside the four quarter-disks of radius R_0. This is an admissible H^1 trial function for the mixed chamber. The four quarter-annuli have total energy $2\pi/\log(R_0/\varepsilon)$, and its squared norm is at least $1 - \pi/16$. Consequently

$$\lambda_1(C_w) \le \frac{2\pi}{(1 - \pi/16)\log(R_0/\varepsilon)} \longrightarrow 0. \tag{9.1}$$

If the four passages have length two, their half-shares assign area $1 + 4w \le 5$ to the chamber. Their first transverse level is π^2/w^2. For small enough ε one can choose an energy below both $4\pi/5$ and π^2/w^2 but above the upper bound in (9.1). The separated chamber count is then at least one, the passage count is zero, and the proposed local area bound is strictly less than

one. Thus that separated comparison really fails. The conclusion concerns the independent cells, not the coupled Dirichlet domain, whose matching conditions were discarded.

At the fixed dimensions of the main construction, all six chamber types are covered, including the fully branching four-port junction. The compact and chain comparisons, together with the explicit nonuniform extension, show how sharp global counts can follow from quantitative boundary corrections, complete tail control, and exact accounting at the actual interfaces. Allowing unrestricted passage widths or lengths requires new local estimates or a comparison that retains more of the coupling at the interfaces. The all-eigenvalue conclusions proved here do not require such an extension.

Appendix A. Exact arithmetic certificates

This appendix specifies the scalar arithmetic, block integrals, and degree-at-most-three and chain event computations used in the proof. All loops have explicit finite bounds, all comparisons use integers or rationals, and all eigenvalue multiplicities are retained.

A.1. **Rational constants.** For a positive integer q, let

$$S_N(q) = \sum_{j=0}^{N-1} \frac{(-1)^j}{(2j+1)q^{2j+1}}.$$

The alternating-series bound and Machin's identity give

$$\ell = 16S_{12}(5) - 4S_5(239) < \pi < 16S_{13}(5) - 4S_4(239) = u.$$

The identity follows by applying the tangent addition formula to $4\arctan(1/5) - \arctan(1/239)$; its tangent is one and its angle lies in $(0, \pi/2)$. Direct rational comparison gives

$$\ell > \frac{103993}{33102} > \frac{333}{106} > \frac{25}{8}, \qquad u < \frac{22}{7}, \qquad u^2 < 10. \tag{A.1}$$

Thus every use of π in the finite checks can be replaced by the displayed rational bounds. Fractions are compared by cross multiplication of their positive denominators.

For completeness, the strict margins in the degree-three boundary calculation reduce to

$$\begin{aligned} A_0 &= \left(\frac{64}{625}\right)^2 \frac{32}{15} + \frac{64}{625}\frac{23}{7}\frac{3}{20} = \frac{2987504}{41015625}, \\ A_0 &< \frac{73}{1000}, \\ \frac{1}{12} + \frac{73}{1000} &= \frac{469}{3000}, \\ 2\left(\frac{73}{1000} + \frac{3}{8}\frac{64}{625}\right) &= \frac{557}{2500}, \\ 4\frac{469}{3000} + \frac{557}{2500} &= \frac{6361}{7500}, \qquad \frac{17}{20} - \frac{6361}{7500} = \frac{7}{3750} > 0. \end{aligned}$$

These give the matrix majorant in (3.21). The strict endpoint in (3.22) follows from $\pi^2 < 10$ and

$$\frac{20}{17 \cdot 10 + 20} = \frac{2}{19}, \qquad \frac{25}{16} + \frac{2}{19} = \frac{507}{304}, \qquad \frac{507}{304} - \frac{5}{3} = \frac{1}{912}, \quad \frac{5}{3} - \frac{53}{32} = \frac{1}{96}.$$

The associated endpoint comparison has margin

$$\frac{53}{32}\frac{333}{106} - \frac{26}{5} = \frac{1}{320} > 0.$$

The first six even-sector numerators, in units of $\pi^2/16$, are $1, 9, 17, 25, 25, 41$; the first three odd-sector numerators are $5, 13, 29$. They are obtained from $16m^2 + (2n+1)^2$ and $4(2m+1)^2 + (2n+1)^2$, respectively. The ranges $0 \le m \le 2$, $0 \le n \le 3$ contain these initial levels and every level at or below $53/2$; positivity of the square terms excludes all other indices. This also checks the multiplicity used in the cluster argument.

The simpler boundary comparisons use

$$1+\frac{22}{7}\,4<64, \qquad \frac{4}{3}-\frac{2(7/4)}{25/8}=\frac{16}{75}, \qquad 63\frac{16}{75}>\frac{88}{7}, \qquad 19\frac{4}{3}>\frac{176}{7},$$

together with $\sqrt{3}<7/4$, $\sqrt{13}<4$, and $\sqrt{10}<4$. They verify the opposite-port and both offset-chain lift thresholds.

A.2. **The fractional-block integral lower sums.** Set $L=8192$ and $D=10^9$ for $(a,K,G)=(1,6,2)$ or $(4,4,4)$. Set $L=32768$ and $D=10^{12}$ for $(a,K,G)=(1,13,3)$ or $(4,7,8)$. For each choice and every integer $0\le j\le GL$, define

$$b_j=\operatorname{isqrt}\left(\left\lfloor\frac{D^2j^2}{a^2L^2+j^2}\right\rfloor\right),$$

where $\operatorname{isqrt}(n)=\lfloor\sqrt{n}\rfloor$ for a nonnegative integer n. Thus

$$\frac{b_j}{D}\le\frac{j}{\sqrt{a^2L^2+j^2}}<\frac{b_j+1}{D}.$$

For $0\le j<GL$, let

$$m_j=\min_{0\le s<K}\sum_{k=0}^{K-1}((s-k)j \bmod L),$$
$$h_j=\max\{0,m_j-C_K\}, \qquad w_j=\max\{0,b_{j+1}-b_j-1\}.$$

The residue in the first line is the unique integer in $\{0,\dots,L-1\}$. Formula (4.1) gives $M_K(j/L)=m_j/L$, and (4.2) bounds the integrand below by h_j/L throughout the slope interval $[j/L,(j+1)/L]$. Under the increasing substitution $x=s/\sqrt{a^2+s^2}$, that interval has x-width at least w_j/D. Consequently

$$A(a,K,G)\ge\frac{1}{KLD}\sum_{j=0}^{GL-1}h_jw_j. \tag{A.2}$$

The integer algorithm is completely specified by

```
parameters = [(1,6,2,8192,10**9), (4,4,4,8192,10**9),
              (1,13,3,32768,10**12), (4,7,8,32768,10**12)]
for (a,K,G,L,D) in parameters:
    C = K*(K-1)//2
    total = 0
    b = 0
    for j in range(G*L):
        t = j+1
        next_b = isqrt((D*D*t*t)//(a*a*L*L+t*t))
        m = min(sum(((s-k)*j) % L for k in range(K))
                for s in range(K))
        # Lower height times lower width, both nonnegative.
        total += max(0,m-C)*max(0,next_b-b-1)
        b = next_b
    return Fraction(total,K*L*D)
```

Here "return" means recording the value for the current triple. In the listed order, the integer totals are

$$12900023357295, \qquad 5549929653480,$$
$$150528062155176292, \qquad 64044999196269018.$$

After cancellation, (A.2) gives exactly the fractions in (4.6) and (4.13). Their comparisons to 13/50, 169/1000, 353/1000, and 279/1000 are strict rational comparisons. Neither square root is evaluated approximately.

The block errors in (4.3) are, respectively,

$$\frac{6^2\cdot 2}{8}+\frac{6}{2}=12,\qquad \frac{4^2\cdot 4}{8}+\frac{4}{2}=10,\qquad \frac{13^2\cdot 3}{8}+\frac{13}{2}=\frac{559}{8},\qquad \frac{7^2\cdot 8}{8}+\frac{7}{2}=\frac{105}{2}.$$

The remaining degree-at-most-three tail arithmetic is

$$\frac{2}{27}<\frac{9}{121},\qquad 2<\frac{100}{49},\qquad \frac{3}{11}\left(\frac{10}{7}+6\right)=\frac{156}{77},\qquad 800<29^2,$$

and

$$-\frac{547}{2000}\,800+\frac{156}{77}\,29+28=-\frac{50838}{385}<0.$$

For degree four,

$$2\frac{353}{1000}+2\frac{279}{1000}-1=\frac{33}{125},\qquad \frac{3}{11}\left(\frac{10}{7}+8\right)=\frac{18}{7},$$
$$1+\frac{559}{8}+2\frac{105}{2}=\frac{1407}{8},\qquad -\frac{33}{125}\,1024+\frac{18}{7}\,32+\frac{1407}{8}=-\frac{85227}{7000}<0.$$

A.3. **Every compact-cell event.** Set $Q = 10240000$. For each of the five triples

$$(s_x, s_y, d) = (2,2,0), (2,1,1), (2,0,2), (1,1,2), (1,0,3),$$

make zero-filled integer arrays $H[0],\dots,H[Q]$ and $R[0],\dots,R[Q]$. A shared array R may be reused for all five types. The following loops generate their complete multiplicities:

```
# Chamber: x = s_x mod 2, y = s_y mod 2, with these minima.
for x in range(s_x,isqrt(Q)+1,2):
    for y in range(s_y,isqrt(Q-x*x)+1,2):
        H[x*x+y*y] += 1
# Corridor: its transverse index y is positive and divisible by 4.
for x in range(0,isqrt(Q)+1):
    for y in range(4,isqrt(Q-x*x)+1,4):
        R[x*x+y*y] += 1
```

The first loop has step two even when $s_x = 2$ or $s_y = 2$, so zero frequencies are present only in the NN coordinates. Each pair of nonnegative squares satisfying the stated threshold lies in exactly one iteration. Thus the bounds are exhaustive.

Set $h = r = 0$. For $p = 0, 1, \dots, Q$, first replace h by $h + H[p]$ and r by $r + R[p]$. If $H[p] + R[p] > 0$, use $c = c_S(p)$ from (5.2) with $N_H = h$ and check

```
# Count all eigenvalues equal to p before making the comparison.
require(h >= c)
U = 2*(h-c) + d*r
require(103993*(2+d)*p - 16*33102*U >= 0)
```

All quantities are integers. Tracking the number of such checks and their minimum margin produces the actual-event table in Section 5. Checking the same margin with $c = 0$ produces exactly the four deficient events reported there.

Also record (h, r) after $p = 2, 20, 26$. At each applicable reset $q = 8/3, 41/2, 53/2$, use the recorded counts after $p = \lfloor q \rfloor$, evaluate $c_S(q) = 0$, and perform the same comparison with q in place of p, now as a rational. The resulting four margins are precisely those in the reset table. This extra instruction is necessary because the corrected upper count can jump when a correction ceases even though the separated spectrum has no eigenvalue there. All starts of corrections are already actual events. The count at Q is checked because the corridor pair $(x, y) = (0, 3200)$ has this event.

The released implementation uses compact arrays only for the individual multiplicities. Their entries cannot exceed the total number of pairs in the enclosing index rectangles, which is less than $3201^2 < 2^{32}$. Every cumulative count, multiplication, and margin comparison uses arbitrary-precision integers. Thus fixed-width overflow cannot affect an acceptance decision.

A.4. **Every offset-chain event.** Use the separate coordinate $X = 32r^2$ from the offset-chain calculation, and set $Q_c = 2880000 = 32 \cdot 300^2$. Initialize an integer array $J[0], \dots, J[Q_c]$ to zero and execute

```
# Separated chamber, including its zero longitudinal mode.
for m in range(1,isqrt(Q_c//8)+1):
    base = 8*m*m
    for n in range(0,isqrt((Q_c-base)//32)+1):
        J[base+32*n*n] += 1
# Corridor, including its zero longitudinal mode.
for m in range(1,isqrt(Q_c//32)+1):
    base = 32*m*m
    for n in range(0,isqrt((Q_c-base)//2)+1):
        J[base+2*n*n] += 1
```

These ranges are exactly the nonnegative solutions of the two spectral inequalities. The chamber indices satisfy $1 \le m \le 600$, $0 \le n \le 300$; the corridor indices satisfy $1 \le m \le 300$, $0 \le n \le 1200$. Initialize $b = 0$, and for each integer $x = 0, \dots, Q_c$ first add $J[x]$ to b. At every x with $J[x] > 0$, and also at $x = 41, 168$, execute

```
# Equality at the left endpoints is included; resets are excluded.
c = int((40 <= x < 41) or (160 <= x < 168))
require(b >= c)
require(999*x - 6784*(b-c) >= 0)
```

This is the rational comparison (8.14). There are 135424 actual jump levels and 135426 checked levels after the two resets are added. The total coarse multiplicity at Q_c is 423767. The only deficient levels before the correction are $X = 40, 160$. The minimum corrected margin is 255 at $X = 41$, with upper count six; the other reset, $X = 168$, has upper count twenty-four and margin 5016. The endpoint Q_c is itself an actual event and is included.

For the analytic chain tail, the twenty intervals in Table 1 are checked without square roots. For its row (a, j, p, q), check

$$[16a^2 + (4j+1)^2]p^2 \ge (4j+1)^2\, 10^6, \qquad [16a^2 + (4j+3)^2]q^2 \le (4j+3)^2\, 10^6,$$

together with the listed order and disjointness of the intervals for each fixed a. Their width numerators sum to 934. The final tail comparison has the exact positive margin

$$\frac{934}{8000} - \frac{1}{60} - \frac{5}{51} = \frac{139}{68000} > 0.$$

As a second arithmetic check, the affine lower bound at $r = 300$ equals $1201/40$, whose square exceeds 800. For $s = 934/8000$, one also has $2s(1201/40) > 8/3$, so the difference $(sr-5)^2 - 8r/3$ is increasing for $r \ge 300$. This verifies the same tail endpoint and monotonicity used in the text.

Appendix B. Exact arithmetic for the four-port chamber

This appendix specifies the finite certificates used in Section 6. Every comparison uses integers or rational intervals. The spectral complement outside each finite square is supplied by (6.7); no numerical eigenvalue approximation enters the argument.

B.1. **Interval matrices.** Let

$$A = 2^{48}, \qquad S_N(q) = \sum_{j=0}^{N-1} \frac{(-1)^j}{(2j+1)q^{2j+1}},$$

and take the rational Machin bounds

$$\ell = 16S_{12}(5) - 4S_5(239), \qquad u = 16S_{13}(5) - 4S_4(239).$$

The alternating-series argument in Appendix A gives $103993/33102 < \ell < \pi < u < 22/7$. Represent a closed interval by its integer endpoints $[L, U]/A$. For a rational $x = p/q$, $q > 0$, use

$$\mathrm{enc}(x) = \left[\left\lfloor \frac{Ap}{q} \right\rfloor, \left\lceil \frac{Ap}{q} \right\rceil\right] /A.$$

Addition and negation operate exactly on endpoints. For a product, form all four products of integer endpoints, divide their minimum and maximum by A, and round respectively down and up. For division by a positive interval, form all four ratios Ax/y of integer endpoints and round their minimum and maximum outwards. Each resulting interval still has denominator A. Thus every individual operation encloses the corresponding exact operation, even when its two arguments are dependent.

Enclose $1/\pi$ by rounding $1/u$ down and $1/\ell$ up. Evaluate (6.5) as a rational constant plus a rational multiple of this interval. The sine of an even integer multiple of $\pi/4$ is obtained from the repeating integer list $0, 1, 0, -1$, and the zero-frequency term contributes $1/2$ to J. Opposite-parity entries are exactly zero. Construct D, W, and $S_e^{(128)}$ from (6.6) and the definitions in Section 6, using interval operations throughout. In particular, the denominator for a high pair (m, n) is $d_m d_n((m^2 + n^2)/4 - e)$, with no omission of zero-index norm factors.

Here is complete loop-level pseudocode for a row (P_*, s) of the table. A block vector is stored as a list of triples (a, b, c) meaning $\sum c\Psi_{ab}$.

```
e = Pstar/16
low = [(m,n) in increasing m, then increasing n:
       m,n >= 0 and m*m+n*n < 4*e]
block = [[(m,n,1)] for (m,n) in low if m*m+n*n >= s]
if Pstar == 41/2:
    block = [[(0,2,1),(2,0,1)], [(1,2,1)], [(2,1,1)]]
k = len(block)
B(i,m,n) = sum over (a,b,c) in block[i] of
    c*((1+(-1)**(a+m))*J(b,n)+(1+(-1)**(b+n))*J(a,m))
D[i,j] = sum(c*B(i,a,b) for (a,b,c) in block[j])
q[j] = (a*a+b*b)/4 for any (a,b,c) in block[j]
norm[j] = sum(c*c*d[a]*d[b] for (a,b,c) in block[j])
W[j] = (e-q[j])*norm[j]
S = zero k by k matrix
for m = 0,...,128:
    for n = 0,...,128:
        if (m*m+n*n)/4 > e:
            v = [B(i,m,n) for i = 0,...,k-1]
            den = d[m]*d[n]*((m*m+n*n)/4-e)
            S[i,j] += v[i]*v[j]/den for 0 <= i <= j < k
mirror the upper triangle of S
T[i,j] = sum(D[i,t]*D[j,t]/W[t] for t = 0,...,k-1)
         - S[i,j] - D[i,j]/3
for t = 0,...,k-1:
    pivot = T[t,t]
    check lower_endpoint(pivot) > 3/4
    for i = t+1,...,k-1:
        for j = i,...,k-1:
            T[i,j] = T[j,i] = T[i,j]-T[i,t]*T[j,t]/pivot
check len(low)-k equals the tabulated cap
```

All list bounds are finite: each index in `low` is at most $\lfloor\sqrt{4e}\rfloor$. The construction also checks that no Neumann mode equals e, that each block column has one energy, and that every division has a strictly positive denominator interval. The loops use exactly the ordering displayed above. Interval symmetric elimination encloses every exact Schur complement by induction;

positive pivots therefore certify positive definiteness of the exact matrix. The common smallest pivot lower endpoint is (6.10). This proves all 28 finite-matrix comparisons by an explicitly reproducible rational computation.

The uniform tail constant used here needs only the rational check

$$4\left[\frac{4}{128(1-(3413/8)/129^2)}+\frac{2}{129^2-3413/8}+\frac{2(22/7)}{127}\right]=\frac{7536117}{23063327}<\frac{1}{3}.$$

Since $129^2-3413/8>127^2$, this dominates $\varepsilon(128,e)$ for every target in the table.

B.2. **The scalar odd–odd certificate.** Use $D_0=10^{14}$ in this subsection only. Each F_n and D_f is a rational constant plus a rational multiple of $1/\pi$. Substitute $[1/u,1/\ell]$ and round outwards to denominator D_0. For odd $n=1,3,\dots,255$, let $[l_n,u_n]/D_0$ be the resulting enclosure for F_n. For an odd high pair define the positive integer

$$v_{mn}=8(m^2+n^2)-163.$$

Its exact Gram contribution is $128(F_m+F_n)^2/v_{mn}$, and an upper bound in units of $1/D_0$ is

$$\left\lceil\frac{128\max\{(l_m+l_n)^2,(u_m+u_n)^2\}}{v_{mn}D_0}\right\rceil. \tag{B.1}$$

The only excluded low pairs are $(1,1),(1,3),(3,1),(3,3)$. Thus the sum of (B.1) has $128^2-4=16380$ terms and equals 29181627575971. The upper enclosure integer for D_f is 21153037117543. The enclosure for the low moment is

$$\frac{5689200238183}{12500000000000}\le b\le\frac{11378400476367}{25000000000000}.$$

For $B=256$, $e=163/32$ the rational tail majorant is

$$\varepsilon_0=4\left[\frac{4}{256(1-(163/8)/257^2)}+\frac{2}{257^2-163/8}+\frac{2(22/7)}{256}\right]=\frac{9516431}{59161648}.$$

Here $257^2-163/8>256^2$. The integer $\lceil\varepsilon_0\,21153037117543\rceil$ is 3402566104473, and bounds the full omitted Gram sum in units of $1/D_0$. Adding the finite and tail bounds, and substituting the lower bound for b, gives exactly

$$\frac{32}{19}\left(\frac{5689200238183}{12500000000000}\right)^2-\frac{29181627575971+3402566104473}{10^{14}}$$
$$=\frac{213752279113583986864148 9}{92773437500000000000000000}>\frac{1}{50}.$$

This supplies all arithmetic in (6.12).

B.3. **The corrected finite interval.** For the interval $0\le P\le 1707$, create integer histograms h_p,r_p initially zero, and execute

```
for m = 0,...,floor(sqrt(1707)/2):
    for n = 0,...,floor(sqrt(1707)/2):
        if 4*(m*m+n*n) <= 1707:
            h[4*(m*m+n*n)] += 1
for m = 0,...,floor(sqrt(1707)):
    for n = 1,...,floor(sqrt(1707)/4):
        if m*m+16*n*n <= 1707:
            r[m*m+16*n*n] += 1
H[p] = sum(h[j] for j = 0,...,p)
R[p] = sum(r[j] for j = 0,...,p)
events = {p: h[p]+r[p] > 0}
events += all 29 cap endpoints, and 1707
for P in sorted(distinct(events)):
    U = min([H[floor(P)]] + [c for (end,c) in caps if P < end])
```

```
    margin = 3*103993*P - 16*33102*(U+2*R[floor(P)])
    check margin >= 0
```

The cap list consists precisely of the table and (6.13). All arithmetic on endpoints is rational. There are 300 distinct checks. At $P = 0$ one obtains $U = R = 0$. Among positive energies the least margin is 291 at $P = 73$, with $U = 19$ and $R = 12$. In particular the reset values themselves are checked after their caps have been removed. This proves (6.15) with inclusive spectral counts.

B.4. **The coarse bridge to the analytic tail.** Put $Q = 2^{24}$. For each of the parameter sets $(a_0, b_0, s_a, s_b) = (0, 0, 2, 2)$ and $(0, 4, 1, 4)$, construct a histogram by the following loops:

```
for a = a0, a0+sa, ... <= isqrt(Q):
    for b = b0, b0+sb, ... <= isqrt(Q-a*a):
        histogram[a*a+b*b] += 1
```

The first histogram is that of H, and the second that of R. Cumulative sums include the event multiplicities before evaluation. At each union event test the integer margin

$$3 \cdot 103993P - 16 \cdot 33102\big(H(P) + 2R(P)\big).$$

There are 1657038 distinct union events, including zero, and the complete list of negative margins is recorded below. The second column in each pair is $B = H(P) + 2R(P)$.

P	B	P	B	P	B
0	1	104	62	425	252
4	3	160	95	436	258
20	14	164	99	468	276
25	16	169	101	596	353
32	19	185	109	601	355
41	25	212	126	833	491
68	41	272	162	1108	654
73	44	292	173	1360	802
80	50	296	175	1361	804
100	60	340	201	1700	1005

All other events have positive margins; the smallest is 7323 at $P = 17$. The endpoint counts are

$$H(Q) = 3296231, \qquad R(Q) = 3292602.$$

The histogram bounds include every possible index because both squared coordinates are nonnegative. Thus this is an exhaustive finite check, including repeated representations and the endpoint Q. The maximum rational catch-up coordinate among the 30 rows is

$$\max \frac{176544B}{103993} = \frac{177426720}{103993} < 1707.$$

This gives the real-energy bridge explained in Section 6. Its endpoint Q agrees with the analytic tail (4.14). The integral certificates behind that tail are already specified by (4.13) and (A.2).

Note on AI Use

The author declares the use of consumer-grade Gemini-3 and GPT-6 models for prior-art search, exploring and developing ideas and proofs, drafting and LaTeX typesetting; the author takes full and sole responsibility for all contents of this work.

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